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REVIEW PAPER

Robot localization: a comprehensive survey from classical methods to intelligent autonomy

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Abstract
Building on previous surveys that often focus on specific components of robot localization, this work offers an integrated review spanning algorithmic estimators, mapping paradigms, and sensor modalities. The survey proposes a functional taxonomy that links probabilistic inference, map representations, and sensor-derived spatial information within a unified comparative framework. By examining classical approaches alongside emerging trends such as semantic mapping and learning-augmented estimators, the paper outlines performance trade-offs, open challenges, and potential directions for future research. Structured tables provide a comparative overview of each class of methods across accuracy, robustness, scalability, and computational cost.

Keywords Robot Localization · SLAM · Sensor Fusion · Kalman Filter · Particle Filter · Large Language Models (LLMs) · Vision-Language Models (VLMs)

1 Introduction

Robot localization is the problem of estimating a robot's pose over time with respect to a known or partially known environment using noisy sensor observations and motion commands. Reliable localization is a prerequisite for almost every non-trivial autonomous behaviour: mobile robots navigating warehouses and hospitals, agricultural platforms operating in unstructured outdoor fields, assistive robots collaborating with humans in homes, and autonomous vehicles sharing the road with human drivers all depend on accurate, robust, and timely estimates of where they are and how they move.

From a systems perspective, localization couples three tightly interacting components. First, *core estimation algorithms* fuse proprioceptive and exteroceptive measurements into a consistent belief over the robot state, typically formulated in $\mathrm{SE}(2)$ or $\mathrm{SE}(3)$ (Sect. 2). Second, *map representations* encode the geometry, topology, and semantics of the surrounding environment at different levels of abstraction (Sect. 3). Third, *sensor modalities* provide the raw cues used for both mapping and state estimation, ranging from wheel encoders, inertial measurement units, and ultra-wideband (UWB) anchors to cameras, LIDAR, and global navigation satellite systems (GNSS) (Sect. 4). Practical localization pipelines emerge from the interplay of these three pillars rather than from any component in isolation.

Despite decades of progress, robot localization remains challenging in both theory and practice. Key difficulties include reliable landmark extraction and data association in cluttered or weakly structured environments; perceptual aliasing, where distinct places generate similar observations; drift accumulation in odometry and visual odometry, and the need for robust loop closure to restore global consistency; limited observability under poor texture, adverse weather, or sparse feature distributions; scalability to large-scale and long-term deployments under real-time and resource constraints; and robustness to dynamic scenes, non-stationary sensor characteristics, and changes in environment appearance. Modern systems must also address challenges related to semantic understanding, explainability, and privacy when operating in human-populated spaces.

Historically, localization research evolved from purely geometric and kinematic reasoning to probabilistic,

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optimization-based, and learning-augmented formulations. Early mobile robots relied on dead-reckoning and geometric triangulation using beacons or landmarks. The introduction of probabilistic state estimation, notably Kalman filtering and Bayesian filtering, allowed principled fusion of noisy motion and measurement models and set the stage for simultaneous localization and mapping (SLAM). Visual SLAM pipelines, reviewed for example by Fuentes-Pacheco et al. (2015), enabled camera-based localization in richly textured environments and motivated tightly coupled visual-inertial systems such as Rosinol et al. (2020). Parallel advances in inertial navigation and multi-sensor fusion provided mature frameworks for integrating IMUs, odometry, and GNSS (Titterton and Weston 2004; Groves 2013), while UWB and related radio-based systems extended reliable positioning to indoor and GPS-denied settings (Al-Okby et al. 2024; Ahmed et al. 2025). More recently, deep learning has been used to learn features, priors, and end-to-end pose regressors, as surveyed by Chen et al. (2023) and in works on deep and semantic SLAM (Zou et al. 2019; Choi et al. 2025).

A rich body of surveys reflects this evolution, but most reviews adopt a perspective centred on a single axis of the problem. Some focus primarily on SLAM as an estimation and optimization problem, often emphasizing either filter-based or graph-based formulations and their scalability properties (Cadena et al. 2016). Others concentrate on specific sensing technologies (e.g., vision, LIDAR, UWB, or GNSS) or application domains (e.g., indoor positioning, autonomous driving, or agriculture), and still others review deep-learning-enhanced SLAM pipelines or semantic mapping methods in isolation (Chen et al. 2023; Zou et al. 2019; Choi et al. 2025). While these surveys provide valuable overviews within their respective scopes, they typically do not develop a unified functional taxonomy that jointly considers estimation algorithms, map representations, and sensing technologies based on their role in spatial reasoning and decision making.

In contrast, this survey adopts a system-level, function-driven view of robot localization. Rather than grouping methods by hardware platform or by narrow algorithmic families, we classify techniques according to *what information they provide to the localization pipeline*: relative motion estimation, globally anchored pose, environmental geometry and topology, or semantic and contextual cues. This perspective highlights how different estimation architectures exploit different map structures and sensor combinations; why certain methods dominate in particular operating regimes (e.g., indoor vs. outdoor, structured vs. unstructured, short-term vs. long-term); and how emerging learning-based and semantic approaches can be composed with classical probabilistic and optimization-based backends. Figure 1 summarizes this organization and illustrates how the three pillars interact across the localization stack, while Fig. 2 sketches the historical progression from early odometry to learning-augmented and language-informed localization systems.

1.1 Contributions of this survey

To clarify the novelty of this review compared with prior surveys, the main contributions are summarized as follows:

- **Unified functional taxonomy across algorithms, maps, and sensors:** We organize the literature around the roles that estimators, map representations, and sensing modalities play in the localization pipeline—relative pose estimation, global anchoring, geometric and

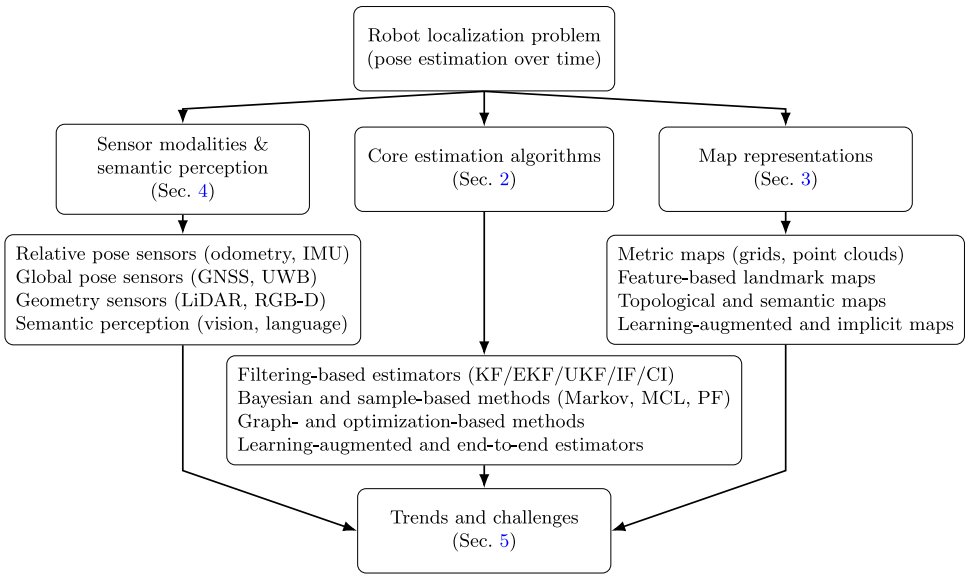


Fig. 1 High-level organization of this survey. The robot localization problem is analyzed through three interacting pillars: core estimation algorithms (Sect. 2), map representations (Sect. 3), and sensor modalities and semantic perception (Sect. 4). Each pillar is structured into functional subcategories, and Sect. 5 synthesizes cross-cutting trends and challenges

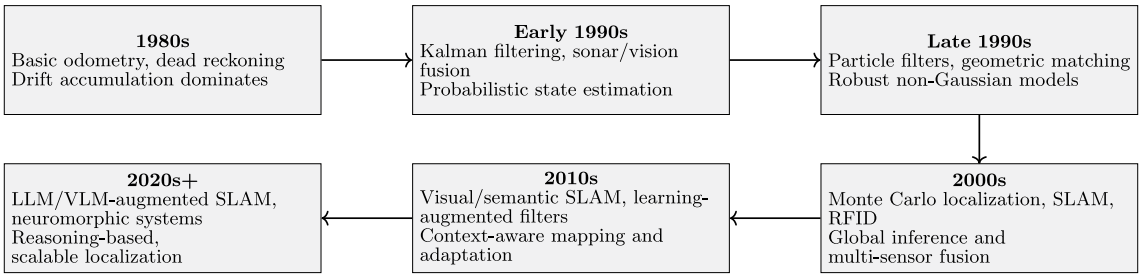


Fig. 2 Evolution of robot localization techniques from early odometry and probabilistic filters to learning-augmented and language-informed localization

Table 1 Unified conventions for qualitative performance levels and table column semantics

Performance level conventions		Table column semantics	
Label	Meaning/Description	Label	Meaning/ Description
L	Low (e.g., low accuracy, limited robustness, low computational cost)	Acc.	Localization accuracy (qualitative levels: L, M, H)
M	Medium (moderate characteristics)	Rob.	Environmental robustness (qualitative levels: L, M, H)
H	High (e.g., high accuracy, strong robustness, high computational cost)	Cost	Computational cost (qualitative levels: L, M, H)
		Scale	Deployment scalability (qualitative levels: L, M, H)

topological structure, and semantic context—rather than around hardware type or single-algorithm families. This yields a taxonomy that is both comprehensive and directly usable for system design.

- **Cross-cutting comparative framework:** Building on this taxonomy, we provide structured comparisons that cut across estimator classes, map types, and sensors, analysing trade-offs in accuracy, robustness, scalability, and computational cost. Where available, we ground qualitative Low/Medium/High ratings with representative quantitative figures drawn from the literature, and we highlight how different combinations of components perform under typical deployment constraints.
- **Semantic and cognitive localization:** We emphasize the transition from purely geometric pipelines to context-aware and semantic localization, with particular focus on the emerging role of LLMs and VLMs as enablers of high-level reasoning, language-guided spatial understanding, and human-robot interaction.
- **Forward-looking perspective on emerging paradigms:** Beyond consolidating established techniques, we identify persistent gaps—including scalability, long-term robustness, generalization, and interpretability—and outline research directions such as learning-augmented filters, differentiable and neuromorphic SLAM, cross-modal localization, and embodied LLM reasoning.
- **Bridging classical methods and intelligent autonomy:** By placing classical probabilistic and optimization-based localization methods alongside modern learning-based, semantic-aware, and multi-sensor approaches in a common framework, this survey provides both an entry point for researchrs that are new to the field and a consolidated reference for practitioners and researchers designing next-generation localization systems.

To enhance accessibility, visual overviews situate individual techniques within the broader localization ecosystem. The conceptual landscape in Fig. 1 highlights the interaction between estimation algorithms, maps, and sensors, while the timeline in Fig. 2 summarizes the historical evolution from dead-reckoning and probabilistic filters to learning-augmented and language-informed localization.

1.2 Notation and conventions

This section establishes the standardized notation and conventions adopted throughout the comparative analyses in this survey. To facilitate consistent interpretation of the diverse robot localization methods, readers are encouraged to consult the following reference tables. Table 1 defines the qualitative performance levels (L, M, H) applied to metrics such as accuracy, robustness, and scalability, and specifies the semantics of the column headers employed across all comparative tables. In addition, Table 9, provided at the end of the paper, lists all technical abbreviations together with their expanded forms to ensure clarity and uniformity of terminology. These conventions are intended to maintain a coherent semantic framework across the subsequent sections.

The comparison tables intentionally use qualitative levels (L/M/H) rather than absolute numerical errors. This choice reflects the strong heterogeneity of the underlying literature: different works evaluate different estimator families on distinct robots, sensor suites, trajectories, and datasets, and they often report different metrics (such as drift percentage,

absolute trajectory error, loop-closure rate, or consistency tests). As a result, a single, unified numeric ranking across all methods would be misleading. Instead, the L/M/H labels summarize typical performance *ranges* reported in representative studies, together with widely accepted order-of-magnitude behavior (for example, calibrated wheel odometry commonly exhibits drift on the order of one to a few percent of the travelled distance, while modern graph-based SLAM can achieve centimetre-level absolute trajectory error on standard benchmarks). Where appropriate, we provide concrete numerical examples in the text, and we explicitly state that the qualitative levels are indicative rather than exact.

2 From kinematics to cognition: core estimation techniques in robot localization

Core estimation algorithms lie at the heart of robot localization: they transform noisy control inputs and sensor measurements into a coherent belief over the robot state, often jointly estimating aspects of the environment. This section traces the evolution of these estimation techniques from early odometry-based dead reckoning to probabilistic filtering, sample-based Bayesian estimators, graph-based smoothing, and learning-augmented architectures. Throughout, we emphasize how design choices relate to the challenges outlined in Sect. 1, such as drift accumulation, perceptual aliasing, limited observability, scalability, and robustness in dynamic environments.

From a functional standpoint, the methods reviewed here answer the question of *how* to fuse information from sensors into consistent pose estimates, given a particular map representation and sensor suite. Section 3 complements this view by examining *what* is represented about the environment (metric, topological, or semantic structure), while Sect. 4 focuses on *which* spatial cues different sensors provide (relative motion, absolute position, geometry, semantics). Practical localization pipelines emerge from the interplay of these three pillars rather than from any component in isolation.

Taxonomic note.

All methods discussed in this section are Bayesian in the probabilistic sense: they maintain or approximate a posterior belief over the robot state given actions and observations. For clarity, however, we group them into four families according to their *representation of uncertainty* and *inference machinery*:

- *Filtering-based estimators* (Sect. 2.2) rely on parametric Gaussian approximations and closed-form or linearized updates, including the Kalman Filter (KF), Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), and information-form variants.
- *Bayesian and sample-based methods* (Sect. 2.3) represent the belief non-parametrically, either on a discrete grid (Markov localization) or via sets of weighted particles (Monte Carlo Localization and generic particle filters).
- *Graph-based and optimization-based methods* (Sect. 2.4) formulate localization as nonlinear least squares on factor or pose graphs, often solving large batches of constraints using sparse optimization.
- *Learning-augmented estimators* (Sect. 2.5) integrate machine learning components (e.g., deep networks) into the estimation pipeline, whether to learn measurement models, priors, or even end-to-end pose regressors.

Separating these families by representation and inference avoids confusion between the general notion of Bayesian estimation and the specific use of the term in the sample-based literature, while clarifying how each class interacts with map representations and sensor modalities.

We begin with odometry and proprioceptive methods, which provide locally consistent but drift-prone relative motion estimates, and then introduce probabilistic filters, sample-based estimators, graph-based optimization, and learning-augmented approaches in roughly chronological order.

2.1 Odometry and proprioceptive methods

Early mobile robots relied almost exclusively on proprioceptive sensing to estimate motion. These methods infer *relative* pose changes from wheel rotations or inertial measurements without explicit reference to external landmarks or maps. Although simple and inexpensive, they are fundamentally limited by drift and unmodelled biases.

Wheel odometry. Wheel odometry integrates wheel encoder counts to estimate linear and angular displacements under kinematic assumptions about the drive configuration (differential, synchro, Ackermann, omni-directional, etc.). Classical work by Borenstein and co-workers (Borenstein and Feng 1996; Borenstein 1995) systematically analysed error sources such as unequal wheel diameters, wheel slip, encoder quantization, and misalignment, and proposed practical calibration procedures and error models. Subsequent studies refined these models and extended them to more complex platforms and terrain conditions (Chong and Kleeman 1997; Kelly 2001; Martinelli 2002; Lee et al. 2012). Wheel odometry offers high-rate, low-latency motion estimates that are cheap and energy-efficient, making it attractive for indoor mobile robots and AGVs.

However, the purely kinematic nature of wheel odometry leads to unbounded error growth in the absence of external corrections. Even with careful calibration, drift on the order of 1–2% of distance travelled is typical for differential-drive platforms operating on nominally flat indoor floors (Borenstein and Feng 1996; Thrun et al. 2005). These characteristics motivate its ubiquitous use as a short-term motion prior that must be regularly corrected by exteroceptive sensors and higher-level estimation algorithms.

Inertial measurement units (IMUs). Inertial measurement units provide linear accelerations and angular velocities at high frequencies (typically 100–1000 Hz), enabling dead-reckoning via strapdown inertial navigation (Titterton and Weston 2004). IMUs capture fast, transient motions that may be poorly sampled by other sensors and are therefore crucial for agile platforms and for bridging short sensor outages. When integrated in isolation, however, inertial navigation systems suffer from rapidly growing drift due to biases, scale errors, and integration noise. High-grade navigation IMUs can bound drift over several minutes, but are expensive and heavy; low-cost MEMS IMUs commonly used in mobile robots drift significantly within seconds (Titterton and Weston 2004; Groves 2013).

Hybrid schemes that combine wheel odometry and IMUs can partly compensate for the weaknesses of each sensor (Lee et al. 2012). For example, angular rates from the IMU can improve heading estimates during wheel slip, while encoders help bound accelerometer bias growth during low-dynamics phases. These proprioceptive fusions are typically embedded within filtering or optimization frameworks, anticipating the more general sensor fusion pipelines discussed in Sects. 2.2 and 2.4 and revisited from a sensing perspective in Sect. 4.

Discussion. Proprioceptive odometry methods are indispensable for providing high-rate motion priors, but they cannot address global consistency or perceptual aliasing on their own. Their error characteristics directly motivate the introduction of probabilistic filters that combine motion models with exteroceptive observations to limit drift and maintain consistent pose estimates over longer time horizons.

2.2 Filtering-based estimators

The limitations of purely proprioceptive methods led to the adoption of probabilistic filters that combine motion and observation models to maintain an estimate of the robot state and its uncertainty over time. Filtering-based estimators represent the belief using a small number of parameters, typically a Gaussian mean and covariance, and update these parameters recursively as new data arrive.

Linear and extended Kalman filtering. The Kalman Filter (KF) (Kalman 1960; Maybeck 1979; Welch and Bishop 2001; Simon 2006) provides an optimal linear-Gaussian solution for state estimation under linear dynamics and observation models with additive Gaussian noise. Although these assumptions are rarely satisfied exactly in mobile robotics, the KF became the foundation for early sensor fusion systems, combining wheel odometry, inertial measurements, and range sensors (Guivant et al. 2002; Hashimoto et al. 1999; Teslić et al. 2010). To handle nonlinear dynamics and observation models, the Extended Kalman Filter (EKF) linearizes the system around the current estimate. The EKF dominated early SLAM work (Bailey and Durrant-Whyte 2006; Thrun et al. 2005), enabling joint estimation of robot pose and landmark positions.

However, linearization can introduce inconsistency, particularly under strong nonlinearities or when the uncertainty is large. The quadratic complexity in the number of landmarks further limits EKF-SLAM scalability, motivating alternative formulations that better exploit sparsity and nonlinearity.

Unscented and other sigma-point filters. Sigma-point filters, such as the Unscented Kalman Filter (UKF) (Wan and Van Der Merwe 2000; Julier and Uhlmann 2004; Bailey and Durrant-Whyte 2006), propagate a set of deterministically chosen sigma points through the nonlinear dynamics and observation models, capturing mean and covariance without explicit Jacobians. These methods often offer improved accuracy over EKF in strongly nonlinear settings while retaining a Gaussian representation. They have been applied, for example, to visual-inertial odometry and tightly coupled GNSS/INS systems (Titterton and Weston 2004; Bader et al. 2017; Erfani et al. 2019).

Information filters and covariance intersection. Information-form filters represent the belief using the information matrix (inverse covariance) and information vector rather than covariance and mean. This can simplify the fusion of conditionally independent measurements and facilitates certain sparsity-exploiting formulations (Maybeck 1979; Bailey and Durrant-Whyte 2006; Borges and Aldon 2003). In multi-sensor or multi-robot settings where cross-covariances are unknown or impractical to maintain, Covariance Intersection (CI) provides a conservative fusion rule that guarantees consistency without requiring explicit correlation models (Bailey and Durrant-Whyte 2006; Carrillo-Arce et al. 2013; Chang et al. 2021; Wang and Liu 2023).

While attractive conceptually, information filters and CI have important limitations. Information-form updates can densify the information matrix when applied naively, eroding sparsity and leading to increased computational cost, particularly in high-dimensional SLAM problems. CI trades away information to preserve consistency, often yielding overly conservative estimates; repeated use can significantly inflate uncertainty and reduce effective information

gain compared with methods that explicitly track correlations (Bailey and Durrant-Whyte 2006; Carrillo-Arce et al. 2013). For these reasons, information-form and CI-based schemes tend to be reserved for specific distributed or communication-constrained scenarios rather than general-purpose SLAM back-ends.

Discussion. Filtering-based estimators remain widely used due to their conceptual simplicity, closed-form recursive updates, and modest computational demands. They are particularly effective for moderate-dimensional state spaces and well-modelled sensors, such as tightly coupled inertial–GNSS navigation. However, the Gaussian approximation struggles with strongly multimodal or highly nonlinear posteriors, and the need to preserve real-time performance in large environments motivates alternative representations. These limitations led to the development of sample-based Bayesian methods and smoothing-based formulations that better capture non-Gaussian uncertainty and exploit sparsity in the underlying estimation problem.

2.3 Bayesian and sample-based methods

To overcome the limitations of parametric Gaussian filters in highly nonlinear, multimodal settings, researchers turned to non-parametric Bayesian estimators that represent the belief explicitly over a discrete state space or via weighted samples.

Grid-based Bayesian estimation (Markov localization). Markov localization (Cox 1990; Koenig and Simmons 1996; Thrun 1998; Thrun et al. 2005) discretizes the robot's state space into a grid and maintains a probability for each cell, updating the distribution with Bayes' rule as actions and observations arrive. Early work combined sonar or laser range data with occupancy grid maps to perform global localization and recovery from severe pose uncertainty, such as the kidnapped robot problem. Grid-based methods naturally handle strong nonlinearities and multimodality (e.g., multiple symmetric hypotheses in highly aliased environments), but they suffer from the curse of dimensionality: the number of grid cells grows exponentially with state dimension and resolution, limiting practical applications to low-dimensional problems or relatively coarse grids.

Particle filters and Monte Carlo Localization. Sequential Monte Carlo methods (particle filters) approximate the posterior by a set of weighted samples drawn from a proposal distribution vGordonspsSalmondspsSmith:1993,DoucetspsDeFreitasspsGordon:2000,ArulampalamspsMaskellspsGordonspsClapp:2002,RisticspsArulampalamspsGordon:2004. Monte Carlo Localization (MCL) (Fox et al. 1998; Thrun et al. 2001, 2005; Dellaert et al. 1999; Lenser and Veloso 2000; Grisetti et al. 2007) applies particle filters to the robot localization problem, representing each pose hypothesis by a particle and updating weights based on sensor likelihoods given a map.

MCL efficiently handles global localization and re-localization after failures, focusing computational effort on high-probability regions of state space. Numerous variants have been proposed, including adaptive resampling and population size control (Fox et al. 1998), mixture MCL and multi-hypothesis schemes for robust global localization (Dellaert et al. 1999; Lenser and Veloso 2000), and Rao–Blackwellized particle filters that factor the problem into low-dimensional trajectory particles with conditionally Gaussian landmark estimates (Thrun et al. 2001; Grisetti et al. 2007). These improvements trade off computational cost, memory, and robustness to perceptual aliasing and sensor noise.

Discussion and relation to filtering. Grid-based and particle-filter methods are Bayesian in the same sense as the Gaussian filters of Sect. 2.2, but they approximate the posterior non-parametrically and can therefore represent complex, multimodal beliefs. In practice, filtering-based estimators remain attractive for tightly constrained, near-Gaussian problems, whereas sample-based methods are preferred for global localization, kidnapped-robot scenarios, and settings with severe perceptual aliasing or highly nonlinear sensors (e.g., range-only localization in cluttered environments) (Thrun et al. 2005; Ristic et al. 2004). The choice of representation is also linked to map structure: occupancy grids and feature maps support efficient likelihood evaluations for particle filters and Markov localization, while certain metric or semantic map types can favour graph-based or learning-augmented approaches, as discussed in Sects. 3 and 2.5.

2.4 Graph-based and optimization-based localization

As environments and trajectories grew larger, the limitations of purely filtering-based formulations became apparent: linearization-induced inconsistency, difficulty revisiting past decisions, and high cost for joint state–map estimation with many landmarks. Graph-based and optimization-based methods address these issues by representing localization (and often mapping) as a sparse nonlinear least-squares problem over a graph.

Pose graphs and factor graphs. In pose graph SLAM, nodes represent robot poses and edges encode relative pose constraints derived from odometry, loop closures, or other sensors (Lu and Milios 1997; Howard et al. 2001; Olson et al. 2006; Grisetti et al. 2010). More generally, factor graphs represent the joint posterior as a product of factors, each relating a small set of variables (poses, landmarks, calibration parameters) (Dellaert 2012). This perspective

unifies a wide range of estimation problems and exposes the sparsity structure that arises from local sensor interactions.

Smoothing versus filtering. Graph-based methods perform *smoothing*: they estimate an entire trajectory (and in SLAM, a map) by minimizing the global error induced by all measurements (Dellaert and Kaess 2006; Bailey and Durrant-Whyte 2006). In contrast to filtering, smoothing allows past poses and landmarks to be refined as new information (e.g., loop closures) becomes available. Incremental solvers such as iSAM and iSAM2 (Kaess et al. 2008, 2012) exploit the sparse structure of the normal equations to update solutions efficiently in an online fashion. Modern SLAM systems routinely combine filtering-style front-ends (e.g., odometry and visual-inertial estimation) with graph-based back-ends for global consistency (Grisetti et al. 2007; Hess et al. 2016; Campos et al. n.d.; Rosinol et al. 2020).

Trade-offs and interaction with maps and sensors. Optimization-based methods offer excellent scalability and consistency for large-scale problems, particularly when combined with sparse linear solvers and careful marginalization strategies. They naturally incorporate heterogeneous constraints from diverse sensors and map types, including metric maps, landmark-based representations, and semantic or object-level information that yields additional factors (Grisetti et al. 2010; Dellaert 2012). However, they generally rely on good initialization (often provided by odometry, visual odometry, or inertial–GNSS filtering) and can be sensitive to outliers, requiring robust cost functions and loop-closure validation.

Section 3 revisits graph-based methods from the perspective of the underlying map representation (e.g., pose graphs versus landmark graphs versus dense maps), while Sect. 4 discusses how specific sensor modalities (LIDAR, cameras, UWB, GNSS) contribute constraints to these optimization problems.

2.5 Learning-augmented localization estimators

Recent advances in machine learning and deep neural networks have motivated a range of learning-augmented localization methods. Rather than replacing classical estimators outright, most successful approaches use learning to enhance specific components: feature extraction, measurement models, data association, or prior representations.

Learned measurement models and priors. A common pattern is to retain a classical filter or graph-based back-end while replacing hand-crafted measurement models or features with learned counterparts. Deep networks can, for instance, predict dense depth maps or optical flow from monocular images, which are then integrated into a visual odometry or SLAM pipeline; or they can produce robust place-recognition descriptors used to generate loop-closure constraints (Walch et al. 2017; Sarlin et al. 2020; Mun et al. 2019). Other work learns priors over trajectories or maps, biasing the estimator toward feasible motions or typical scene layouts (Gellert and Schlögl 2018; Dourado et al. 2019).

End-to-end pose regression. End-to-end approaches such as PoseNet (Kendall et al. 2016) train convolutional or transformer-based networks to regress camera pose directly from images, bypassing explicit feature extraction and geometric optimization. Variants incorporate temporal models, uncertainty estimation, and multi-task learning (Ching-Chih et al. 2015; Xiao et al. 2019; Suzuki 2019). While these methods can perform well in constrained environments and offer appealing deployment simplicity, they often struggle to generalize beyond the training distribution and provide limited interpretability compared with classical geometry-based pipelines.

Hybrid and semantic-aware estimators. Hybrid systems combine probabilistic estimators with learned components that provide semantic or contextual information. For instance, semantic segmentation networks can label objects and surfaces in the environment, and these labels can be incorporated into the map representation or used to modulate sensor models, improving robustness under appearance changes and dynamic objects (Zou et al. 2019; Chen et al. 2023). Recent work explores integrating language-conditioned priors and high-level reasoning, where LLMs and VLMs contribute semantic constraints or task-level context to the localization process (Choi et al. 2025). These trends blur the line between purely geometric localization and more cognitive forms of spatial understanding.

Discussion. Learning-augmented estimators hold promise for improving robustness, adaptability, and semantic awareness, particularly in challenging environments where hand-crafted models are brittle. At the same time, they introduce new challenges related to data requirements, generalization, uncertainty quantification, and safety. In this survey, we treat learning-augmented methods as complementary to classical probabilistic and optimization-based techniques: they enrich the information available to the estimator and can modify its behaviour, but they rarely obviate the need for principled state estimation. Section 3 further discusses how learned representations influence map design, while Sects. 4 and 5 revisit learning-based sensing and reasoning in the broader context of future localization systems.

2.6 Summary and future directions

Tables 2 and 3 summarize the core estimator families reviewed in this section from a functional perspective, using the qualitative conventions introduced in Table 1. Rows correspond to the main estimator classes, while columns

Table 2 Summary comparison of core estimator families in robot localization. Qualitative levels (L, M, H) follow the conventions in Table 1.

Estimator family	Acc.	Rob.	Cost	Scale
Odometry and proprioceptive methods	M	L–M	L	M
Filtering-based estimators (KF/EKF/UKF/IF/CI)	M–H	M	M	M
Bayesian and sample-based methods (Markov, MCL, PF)	M–H	H	H	M
Graph-based and optimization-based methods	H	H	M–H	H
Learning-augmented estimators (learned models, hybrids, end-to-end)	M–H	M–H	M–H	M

capture typical trade-offs in accuracy, robustness, computational cost, and scalability, as well as the map structures and sensor modalities they are most commonly paired with. These ratings are indicative and based on representative results reported in the literature; for example, wheel odometry alone often exhibits drift on the order of 1–2% of distance travelled indoors (Borenstein and Feng 1996; Thrun et al. 2005), whereas modern graph-based SLAM systems routinely achieve centimetre-level consistency over hundreds of metres given reliable loop closures and LIDAR or camera data (Grisetti et al. 2010; Campos et al. 2021).

From this summary, several trends and open questions emerge. First, there is a clear progression from locally valid, drift-prone estimators (odometry and pure proprioception) toward globally consistent, multi-sensor architectures (graph-based and learning-augmented methods) that explicitly exploit map structure and semantic context. Second, robustness to perceptual aliasing and dynamic environments remains challenging: particle methods and graph-based SLAM can represent multimodal hypotheses and incorporate robust cost functions, but they rely on reliable data association and loop-closure validation, which are often brittle in highly dynamic or weakly structured scenes. Third, learning-based components offer powerful feature representations and semantic cues, yet raise concerns about generalization beyond the training distribution, explainability, and safety-critical uncertainty quantification.

Estimator choice in typical scenarios. From a system-design perspective, different estimator families tend to be preferred in different operating regimes:

Table 3 Unified comparison of estimator families in terms of typical strengths and limitations

Estimator family (representative works)	Typical strengths	Typical limitations
Odometry and proprioceptive methods (Borenstein and Feng 1996; Borenstein 1995; Chong and Kleeman 1997; Kelly 2001; Martinelli 2002; Lee et al. 2012; Titterton and Weston 2004)	Simple, low-cost hardware; high-rate motion estimates; low computational cost; suitable as a motion prior and for short-term dead reckoning	Unbounded drift; sensitive to wheel slip, terrain irregularities, and sensor biases; no direct global localization; must be corrected by exteroceptive sensing
Filtering-based estimators (KF/EKF/UKF/IF/CI) (Kalman 1960; Maybeck 1979; Welch and Bishop 2001; Simon 2006; Guivant et al. 2002; Hashimoto et al. 1999; Teslić et al. 2010; Wan and Van Der Merwe 2000; Bailey and Durrant-Whyte 2006; Julier and Uhlmann 2004; Borges and Aldon 2003; Carrillo-Arce et al. 2013; Chang et al. 2021; Wang and Liu 2023)	Principled Bayesian fusion of motion and measurements under Gaussian assumptions; real-time capable; well-understood tuning; effective for tightly coupled inertial–GNSS and moderate-scale SLAM	Gaussian approximation struggles with strong nonlinearity and multi-modality; EKF can become inconsistent; scalability issues for large landmark sets; information-form and CI variants may become dense or overly conservative
Bayesian and sample-based methods (Markov localization, MCL, PF) (Cox 1990; Koenig and Simmons 1996; Thrun 1998; Thrun et al. 2005; Fox et al. 1998; Thrun et al. 2001; Dellaert et al. 1999; Lenser and Veloso 2000; Grisetti et al. 2007; Gordon et al. n.d.; Doucet et al. 2001; Arulampalam et al. 2002; Ristic et al. 2004)	Non-parametric representation of the posterior; naturally handles multi-modality and severe perceptual aliasing; supports global localization and recovery from failure	Computational and memory cost scale with state dimension and particle count; prone to particle depletion; performance dependent on proposal distribution and sensor model design; grid-based methods suffer from the curse of dimensionality
Graph-based and optimization-based methods (Lu and Milios 1997; Howard et al. 2001; Olson et al. 2006; Grisetti et al. 2010; Dellaert and Kaess 2006; Kaess et al. 2008, 2012; Hess et al. 2016; Campos et al. n.d.; Rosinol et al. 2020; Dellaert 2012)	Exploit sparsity for large-scale, globally consistent estimation; refine past poses and landmarks via smoothing; naturally integrate heterogeneous constraints from many sensors and map structures	Require good initialization; sensitive to outliers and loop-closure errors; complexity depends on graph size and connectivity; need robust front-ends for data association and loop-closure validation
Learning-augmented estimators (learned models, hybrids, end-to-end) (Kendall et al. 2016; Ching-Chih et al. 2015; Xiao et al. 2019; Suzuki 2019; Walch et al. 2017; Sarlin et al. 2020; Mun et al. 2019; Gellert and Schlögl 2018; Dourado et al. 2019; Zou et al. 2019; Chen et al. 2023; Choi et al. 2025)	Leverage powerful learned features, priors, and semantic cues; can improve robustness under appearance change and partial observability; enable language-conditioned and task-aware localization	Require substantial training data; generalization beyond the training distribution is not guaranteed; uncertainty quantification and safety certification remain challenging; end-to-end approaches may be difficult to interpret and debug

- *Short-range indoor navigation with limited sensing.* For small service robots equipped with wheel encoders and a 2D LIDAR or depth camera in structured indoor environments, wheel odometry combined with an EKF or UKF and a modest-sized occupancy or feature map is often sufficient. Global localization and recovery from failure are typically handled by Monte Carlo Localization on the same map.
- *Large-scale mapping in predominantly static environments.* For long trajectories in warehouses, office buildings, or urban outdoor scenes, graph-based and optimization-based methods become preferable. Visual, LIDAR, or visual–inertial odometry provides a high-rate front-end, while a pose-graph or factor-graph back-end fuses loop closures, GNSS and UWB anchors, and other constraints to maintain global consistency over hundreds of metres or kilometres.
- *GPS-denied, weakly structured, or highly aliased settings.* In subterranean, forested, or highly repetitive environments where GNSS is unavailable and perceptual aliasing is severe, sample-based methods and robust graph-based formulations are attractive. Particle filters and MCL support global localization and multi-hypothesis tracking, while robust optimization with outlier-resistant costs can downweight spurious loop closures.
- *Semantically rich, dynamic, or human-populated environments.* When operation depends on understanding dynamic objects, semantics, or human activities (e.g., domestic, industrial, or social robotics), learning-augmented estimators that incorporate semantic perception, learned measurement models, or language-conditioned priors become increasingly useful. In practice, these are most effective when coupled to a classical filter or graph-based back-end that enforces physical consistency and provides calibrated uncertainty.

These examples are not prescriptive recipes, but they illustrate how the qualitative trade-offs in Tables 2 and 3 translate into concrete design choices for different application domains and sensing configurations.

Future research on core estimators is likely to focus on tighter integration between principled probabilistic inference and learned models, for example through differentiable filtering and smoothing, uncertainty-aware deep measurement models, and architectures that expose intermediate representations for introspection and verification. Another promising direction is cross-modal and cross-embodiment localization, where estimators leverage shared semantic and geometric structure across different sensor suites and robot platforms. Finally, there is growing interest in using large (vision-)language models as high-level reasoning modules that can interact with classical estimators, for instance by proposing semantic loop closures, querying map inconsistencies, or incorporating human instructions as soft constraints; Sect. 5 revisits these themes in the broader context of trends and open challenges in robot localization.

3 Building the world: map representations in localization

Localization cannot occur in a vacuum: it must be grounded in some representation of the environment. In this survey, *map representations* are understood as any spatial structures that allow an estimator to relate sensor measurements and control inputs to a consistent notion of place. These maps may encode metric geometry, topological connectivity, semantic labels, or learned latent structure. Unlike pure mapping surveys, our focus here is not on map construction alone, but on how different map structures support the *localization* algorithms discussed in Sect. 2 and interact with sensor modalities in Sect. 4.

From a functional point of view, map-based localization answers the question of *where am I within a given representation of the world?* Metric maps (e.g., occupancy grids, point clouds, dense depth maps) support geometric alignment of new measurements to a reference frame. Feature and keyframe maps support correspondence-based and appearance-based localization. Topological and semantic maps provide a more abstract notion of place, useful for large-scale navigation and task-level reasoning. Learning-augmented and implicit representations further extend these ideas by embedding geometry and semantics into neural networks or other learned models.

Taxonomic note. In line with the taxonomy introduced in Sect. 2, we group map-based localization methods according to the *dominant spatial abstraction* used by the estimator:

- *Laser-based metric map localization* (Sect. 3.1) operates on dense geometric representations such as occupancy grids, height maps, or point clouds, typically derived from 2D or 3D LIDAR scans.
- *Vision-based geometric and feature-based localization* (Sect. 3.2) uses sparse or semi-dense feature maps, keyframe graphs, or direct image-to-map alignment based on cameras and depth sensors.
- *Topological and semantic map-based localization* (Sect. 3.3) leverages place graphs, appearance-based indices, and semantic or object-centric maps to reason about locations at a higher level of abstraction.
- *Learning-augmented and implicit map representations* (Sect. 3.4) integrate neural or other learned representations into the map structure itself, often combining geometry and semantics in a common latent space.

Each category is associated with characteristic estimator families and sensor configurations. For example, laser-based metric maps often feed scan-matching front-ends and graph-based back-ends, while semantic maps are frequently paired with learning-augmented estimators and higher-level reasoning modules. Section 3.5 summarizes these relations and links them to the estimator taxonomy of Sect. 2.

3.1 Laser-based metric map localization

Laser-based localization techniques rely on aligning LIDAR scans against a metric representation of the environment, such as an occupancy grid, polygonal map, or point cloud. These methods exploit the high range accuracy and angular resolution of modern LIDAR sensors and remain a cornerstone of many commercial autonomous systems.

A foundational approach is the Iterative Closest Point (ICP) algorithm (Besl and McKay 1992), which iteratively matches points between a new scan and a reference point cloud to recover the rigid transform that minimizes alignment error. Variants such as point-to-plane ICP and generalized ICP (Segal et al. 2009) improve convergence and robustness by incorporating surface normals and probabilistic error models. In parallel, correlative scan matching (Olson 2009) aligns scans by evaluating discrete pose hypotheses over a search window using precomputed occupancy or likelihood grids, trading computation for stronger robustness to local minima.

An alternative formulation, the Normal Distributions Transform (NDT) (Biber and Strasser 2003; Magnusson 2009), represents the map as a set of Gaussian distributions over grid cells. Localization then amounts to maximizing the likelihood of the observed scan under this continuous approximation. Multi-resolution and adaptive-grid variants (Hornung et al. 2013) improve robustness in sparse, noisy, or large-scale environments by focusing computation on informative regions.

In practice, laser-based metric map localization typically appears as a *front-end* that provides relative pose constraints to a filtering or graph-based *back-end* (Sect. 2). Scan-to-map alignment yields odometric constraints, while loop closures arise from aligning scans in revisited areas. The choice of map representation (e.g., occupancy grid versus NDT versus point cloud) influences both accuracy and computational cost, as well as sensitivity to dynamic objects and structural changes.

Overall, laser-based metric methods offer high localization accuracy and strong robustness in static or moderately dynamic environments, particularly when dense geometry is available. Their main limitations are the reliance on sufficient geometric structure, reduced performance in highly dynamic scenes, and non-trivial computational demands for global alignment in large maps.

3.2 Vision-based geometric and feature-based localization

Vision-based localization techniques operate on maps derived from cameras and, increasingly, RGB-D or stereo sensors. These maps typically encode sparse visual features, semi-dense photometric information, or dense depth, organized as keyframes and landmarks in a graph structure. Compared with LIDAR, cameras are inexpensive and provide rich texture and semantic information, but they are more sensitive to lighting changes, motion blur, and textureless surfaces.

Classical feature-based visual localization constructs a sparse landmark map by detecting and matching keypoints (e.g., SIFT, SURF, ORB) across images and estimating camera poses that are consistent with these correspondences (Scaramuzza et al. 2006; Matsuo et al. 2014). The resulting maps are well suited for incremental SLAM and revisiting previously seen places via appearance-based loop closure (Cummins and Newman 2008; Galvez-López and Tardós 2012). Geometric consistency checks and robust estimators (e.g., RANSAC, bundle adjustment) provide resilience against outliers.

Direct and semi-direct visual methods dispense with explicit feature descriptors and instead minimize photometric error between images or between images and a stored keyframe map (Newcombe et al. 2011; Engel et al. 2014). These approaches exploit more of the image information and can be more robust in low-texture settings, but they depend on stable illumination and accurate photometric calibration. Depth cameras and stereo rigs further augment the map with dense or semi-dense geometry, improving scale observability and enabling more accurate alignment.

Vision-based maps are tightly coupled with the estimator design. For instance, keyframe–landmark maps naturally support graph-based SLAM back-ends that optimize camera poses and landmark positions, while semi-dense maps can be used as implicit surfaces for camera tracking. Cameras are also central to semantic and learning-augmented maps (Sects. 3.3 and 3.4), where visual features are combined with higher-level labels and learned representations.

Compared with laser-based metric maps, vision-based representations offer lower sensor cost and richer semantics but are more prone to appearance changes and may provide weaker geometric constraints in texture-poor environments. Hybrid LIDAR–vision maps seek to combine the strengths of both, using LIDAR for robust geometry and vision for long-range appearance-based loop closure and semantics.

3.3 Topological and semantic map-based localization

Topological and semantic maps abstract away from precise metric geometry to represent the environment as a graph of places and their relationships. Nodes correspond to locations, regions, or semantic entities (e.g., rooms, corridors, objects), and edges encode navigability, adjacency, or logical relations. Localization then amounts to inferring the most likely node or set of nodes given current sensor observations.

Appearance-based topological maps represent each place using visual or LIDAR-derived descriptors and organize them in a retrieval structure for loop closure and place recognition (Cummins and Newman 2008; Galvez-López and Tardós 2012). These maps scale gracefully to large environments because they store only a subset of keyframes or descriptors and can employ efficient indexing. Metric information is often maintained in a separate layer (e.g., a pose graph) while the topological layer captures long-range connectivity, enabling robust global localization and multi-session mapping.

Semantic maps enrich metric or topological structures with labels and object instances inferred from perception (Mccormac et al. 2017; Bowman et al. 2017). For example, SemanticFusion and related approaches couple dense SLAM with semantic segmentation networks to build maps where each surface element is associated with a semantic class. Other works build object-centric or room-centric maps that capture the layout of furniture, walls, and functional areas. In these settings, localization can exploit both geometry and semantics, improving robustness under appearance change (e.g., lighting, weather) and supporting higher-level reasoning such as "in the kitchen near the fridge".

Topological and semantic map-based localization integrates naturally with the estimator families of Sect. 2. Graph-based back-ends can treat places and semantic entities as additional nodes with factors that encode co-occurrence or relative relationships. Particle filters can maintain multi-hypothesis beliefs over topological nodes in ambiguous environments, while learning-augmented estimators can provide semantic descriptors and language-conditioned priors that connect human instructions to locations.

The key advantages of these representations are scalability, interpretability, and robustness to certain types of appearance change. Their limitations include reduced metric precision (unless combined with a metric layer), sensitivity to semantic prediction errors, and the need for substantial training data when deep semantic models are used.

3.4 Learning-augmented and implicit map representations

Learning-augmented map representations incorporate neural or other learned models into the map structure itself. Rather than storing geometry and semantics as explicit grids or graphs alone, these methods represent the environment via neural fields, latent codes, or parameterized functions that are optimized from data.

Implicit neural representations, such as neural occupancy fields and radiance fields (NeRF), model the environment as a continuous function mapping coordinates (and possibly viewing directions) to density, color, or semantic labels (Mildenhall et al. 2020; Mescheder et al. 2019). Localization can then be framed as finding the pose whose rendered view best matches the current observation. While these approaches yield high-fidelity reconstructions and compact representations of complex scenes, they often require substantial training time and can be computationally intensive at inference, limiting their current use to offline or slowly changing environments.

Other learning-augmented maps use neural networks to compress or regularize classical structures. For example, learned place-recognition descriptors provide a semantic- and appearance-aware index over keyframes or LIDAR scans (Sarlin et al. 2020; Walch et al. 2017), effectively functioning as a topological layer atop a metric map. Neural networks can also predict occupancy or traversability from raw sensor data, building semantic occupancy grids that better capture free space and dynamic objects than classical inverse sensor models.

These representations are closely intertwined with the learning-augmented estimators of Sect. 2.5: the same network that provides a map can also serve as a learned measurement model or prior. The resulting systems blur the distinction between mapping and localization, as the map is no longer a static structure but part of a learned function that supports both tasks. This flexibility comes at the cost of data requirements, generalization challenges, and difficulties in quantifying uncertainty and ensuring safety-critical robustness.

3.5 Summary and future directions

Tables 4 and 5 summarize the map-based localization categories reviewed in this section using the qualitative conventions from Table 1. The first table compares the categories along accuracy, robustness, computational cost, and scalability; the second highlights typical strengths and limitations, including how each representation interacts with estimator families and sensor modalities.

Table 4 Summary comparison of map-based localization categories. Qualitative levels (L, M, H) follow the conventions in Table 1.

Category	Acc.	Rob.	Cost	Scale
Laser-based metric map localization	H	H	M–H	M–H
Vision-based geometric and feature-based localization	M–H	M	M	H
Topological and semantic map-based localization	M	H	L–M	H
Learning-augmented and implicit map representations	M–H	M–H	M–H	M

Map choice in typical scenarios. From a system-design perspective, different map structures tend to be preferred in different operating regimes:

- *Compact indoor navigation with reliable range sensing.* For mobile robots operating in relatively static corridors and rooms equipped with 2D LIDAR or RGB-D cameras, occupancy grids or NDT-style metric maps are often sufficient. Scan matching and ICP-based front-ends, combined with graph-based back-ends (Sect. 2.4), provide accurate pose estimates with modest memory and computational requirements.
- *Large-scale multi-floor buildings or campuses.* When environments span multiple floors, wings, or buildings, a purely metric map can become unwieldy. Hybrid metric–topological maps, where local metric submaps are connected by a place graph, strike a balance between geometric fidelity and scalability. Appearance-based place recognition from cameras or LIDAR descriptors supports loop closure and multi-session mapping at the topological level.
- *Long-term operation with frequent layout changes.* In warehouses, retail spaces, or offices where furniture or shelving changes over time, maintaining a single consistent metric map is challenging. Topological and semantic maps, which represent places and functional areas (e.g., aisles, storage zones, rooms) more abstractly, are more robust to small geometric changes and can be updated incrementally without rebuilding the entire metric map.
- *Semantics-critical tasks and human–robot interaction.* For service robots that must interpret human instructions ("go to the kitchen and fetch the mug") or reason about objects and activities, semantic and object-centric maps become essential. Here, metric maps provide the geometric substrate for navigation, while semantic layers encode rooms, objects, and affordances. Learning-augmented and implicit representations further support rich queries and language-conditioned reasoning, but typically operate in tandem with classical metric and topological layers.

Design implications and cross-links. From a system-design perspective, metric LIDAR maps are often preferred in applications that demand high geometric accuracy and safety, such as warehouse automation or autonomous

Table 5 Unified comparison of map-based localization categories in terms of typical strengths and limitations. Representative works are indicated in parentheses

Category (representative works)	Typical strengths	Typical limitations
Laser-based metric map localization (Besl and McKay 1992; Segal et al. 2009; Olson 2009; Biber and Strasser 2003; Magnusson 2009; Hornung et al. 2013)	High metric accuracy and repeatability in structured environments; robust scan alignment in static or moderately dynamic scenes; well-suited as a front-end for graph-based SLAM and for safety-critical navigation	Requires sufficient geometric structure and relatively dense scans; performance degrades with severe dynamics, occlusion, or sparse observations; global matching and multi-resolution search can be computationally demanding for very large maps
Vision-based geometric and feature-based localization (Scaramuzza et al. 2006; Matsuo et al. 2014; Cummins and Newman 2008; Galvez-López and Tardós 2012; Newcombe et al. 2011; Engel et al. 2014)	Uses low-cost, ubiquitous sensors; provides rich texture and semantic cues; supports both sparse feature and direct photometric alignment; naturally integrates with graph-based estimators and learning-based place recognition	Sensitive to illumination changes, motion blur, and textureless regions; scale observability can be problematic with monocular setups; mapping quality depends on calibration and feature/descriptor robustness
Topological and semantic map-based localization (Cummins and Newman 2008; Galvez-López and Tardós 2012; Mccormac et al. 2017; Bowman et al. 2017)	Scales to large environments through compact place and semantic abstractions; supports robust global localization and loop closure; offers interpretable, task-relevant representations well-suited for planning and human–robot interaction	Provides limited metric precision unless combined with a metric layer; depends on reliable place-recognition and semantic prediction; semantic models require training data and may not generalize to unseen categories or domains
Learning-augmented and implicit map representations (Mildenhall et al. 2020; Mescheder et al. 2019; Sarlin et al. 2020; Walch et al. 2017)	Capture complex geometry and appearance in compact latent or functional forms; enable joint reasoning about geometry and semantics; integrate naturally with learning-augmented estimators and vision-language models; can be tailored to specific tasks (e.g., traversability)	Training and inference can be computationally expensive; generalization beyond the training distribution is not guaranteed; uncertainty quantification and safety certification are challenging; toolchains are still maturing for real-time, large-scale deployment

driving, especially when combined with graph-based estimators from Sect. 2 and LIDAR modalities from Sect. 4. Vision-based maps are attractive when cost, size, or semantic richness are critical, and they naturally pair with visual(-inertial) odometry front-ends and learning-augmented loop closure. Topological and semantic maps become indispensable for long-term, large-scale deployments and task-level navigation, where they provide a compact, interpretable substrate for localization and planning. Learning-augmented and implicit maps, finally, open opportunities for tighter integration between mapping, localization, and high-level reasoning, but currently require careful validation and hybridization with more classical structures.

These trends suggest that future localization systems will increasingly rely on *multi-layer maps* that combine metric, topological, and semantic representations, with different estimator families operating on different layers. Such systems will need principled mechanisms to propagate uncertainty and information across layers, and to reconcile classical geometric constraints with learned and language-informed priors—a theme revisited in Sect. 5.

4 From sensors to situational awareness: a functional view of localization modalities

Localization is ultimately constrained by what the robot can sense. While Sects. 2 and 3 focused on how estimation algorithms and map representations transform sensor data into pose estimates and spatial models, this section examines the complementary question of *what* spatial information different sensor modalities provide, *how* it is acquired, and *for which roles* in the localization pipeline. Rather than grouping sensors purely by hardware type, we adopt a functional taxonomy based on the spatial cues they contribute: relative motion increments, globally anchored pose, environmental geometry, and semantic or contextual information.

Accordingly, we distinguish four broad categories. Relative pose sensors (Sect. 4.1) provide high-rate egomotion increments but suffer from drift. Global positioning sensors (Sect. 4.2) deliver absolute pose anchoring where infrastructure or satellite coverage is available. Environmental geometry sensors (Sect. 4.3) reconstruct the surrounding structure to support map-based localization and loop closure. Finally, semantic perception modules (Sect. 4.4) infer higher-level information from raw signals, such as object classes or room types, and thus contribute contextual cues rather than directly measuring new physical quantities. Section 4.5 summarizes the resulting trade-offs and relates these modalities back to the estimator and map taxonomies in Sects. 2 and 3.

4.1 Relative pose estimation sensors

Relative pose estimation sensors provide continuous, high-rate estimates of local motion by integrating incremental measurements over time. They are indispensable for short-term stability, control, and bridging between sparser global or map-based updates, particularly when external references such as GNSS or pre-built maps are unavailable or intermittent. However, by construction they cannot recover absolute pose and suffer from unbounded drift if not regularly corrected.

Wheel encoders remain one of the most widely used relative sensors for indoor and ground-based robots. By integrating wheel rotation under a kinematic model of the drive configuration, wheel odometry provides inexpensive, low-latency estimates of planar motion (Borenstein and Feng 1996; Borenstein 1995). As discussed in Sect. 2.1, systematic errors (unequal wheel diameters, misalignment) and non-systematic effects (wheel slip, surface irregularities) lead to drift on the order of 1–2% of distance travelled even under favourable conditions (Borenstein and Feng 1996; Thrun et al. 2005). In the sensor taxonomy, encoders are therefore best seen as high-rate, local motion cues that must be fused with exteroceptive information using the estimators from Sect. 2 and maps from Sect. 3.

Inertial measurement units (IMUs) provide linear accelerations and angular velocities at rates of up to several hundred hertz, enabling strapdown inertial navigation (Titterton and Weston 2004; Groves 2013). High-grade IMUs can maintain accurate short-term motion estimates even under aggressive manoeuvres, making them crucial for aerial vehicles and legged robots (Aggarwal et al. 2008; Shin 2005). Low-cost MEMS IMUs commonly used in mobile robots, however, exhibit substantial bias drift and noise, causing position errors to grow rapidly when integrated in isolation (Groves 2013; Mohamed et al. 2019). In practice, IMUs serve as complementary high-dynamics sensors whose biases and drift are continuously corrected using other modalities within filtering or optimization frameworks.

Visual odometry (VO) estimates relative motion from image sequences by tracking features or minimizing photometric error (Scaramuzza and Fraundorfer 2011). Feature-based methods detect and match keypoints between frames and use multi-view geometry to recover camera pose (Mur-Artal and Tardos 2017), while direct and semi-direct methods operate directly on image intensities (Engel et al. 2014). VO provides rich motion cues using inexpensive cameras and is particularly attractive in GNSS-denied environments. However, it remains sensitive to illumination changes, motion blur, and texture-poor scenes, and, like other relative modalities, accumulates drift over time.

Visual–inertial odometry (VIO) tightly or loosely couples VO with inertial measurements to exploit complementary strengths (Mourikis and Roumeliotis 2007; Leutenegger et al. 2015; Qin et al. 2018). IMU data stabilize visual tracking during rapid motions and provide short-term scale information, while visual observations bound inertial drift and correct biases. Modern VIO systems often form the front-end of visual–inertial SLAM pipelines, with graph-based or filtering back-ends (Sect. 2.4) maintaining global consistency.

In summary, relative pose sensors are crucial for maintaining continuous egomotion estimates and stabilizing control loops, but their drift characteristics mean they are rarely sufficient for long-term localization on their own. They are most effective when paired with global positioning, geometric maps, or semantic perception modules that provide absolute or loop-closing information.

4.2 Global positioning sensors

Global positioning sensors provide absolute pose estimates by relating local measurements to external references anchored in a global frame. They are particularly valuable for large-scale outdoor navigation, multi-robot coordination, and as global priors for SLAM and map-building.

Global Navigation Satellite Systems (GNSS), including the United States' Global Positioning System (GPS), Russia's GLObal NAvigation Satellite System (GLONASS), Europe's Galileo, and China's BeiDou constellations determine receiver position by trilateration from pseudorange measurements to multiple satellites (Groves 2013). Standard code-based receivers typically provide meter-level accuracy in open-sky conditions. Carrier-phase techniques such as Real-Time Kinematic (RTK) and Precise Point Positioning (PPP) achieve centimetre-level accuracy by resolving integer ambiguities using one or more reference stations (Groves 2013; Ahmed et al. 2025). However, GNSS performance degrades in urban canyons, tunnels, forests, and indoor environments due to signal blockage, multipath, and ionospheric effects.

Ultra-wideband (UWB) radio systems offer an attractive complement to GNSS in indoor and GPS-denied settings (Alarifi et al. 2016; Al-Okby et al. 2024). By measuring time-of-flight or time-difference-of-arrival between mobile tags and fixed anchors, UWB can provide decimetre-level ranging and localization even in cluttered environments (Zhao et al. 2020). UWB is robust to many forms of multipath and interference, but requires infrastructure deployment and careful calibration of anchor positions.

In indoor environments where GNSS signals are unavailable or severely attenuated, infrastructure-based systems can provide absolute position anchors within a bounded workspace. Radio-frequency identification (RFID) tags, for example, allow robots to obtain discrete but globally referenced pose information when passing near tagged locations or gateways. Such "indoor GPS" systems are attractive in structured warehouses, factories, and logistics hubs where installing and maintaining the required infrastructure is feasible, but they remain difficult to scale to uninstrumented or dynamically changing environments.

In practice, GNSS and UWB are often fused with relative and geometric modalities in filters or factor graphs. GNSS provides a weak but globally consistent prior that constrains drift in vehicle-scale SLAM (Ahmed et al. 2025), while UWB anchors assist with global localization and robustness in warehouses, factories, and indoor drone applications (Al-Okby et al. 2024). Their main limitations arise from coverage gaps, susceptibility to environmental conditions, and dependence on infrastructure, but when available they significantly simplify the global localization problem by providing absolute anchors.

4.3 Environmental geometry sensors

Environmental geometry sensors enable robots to perceive the shape and structure of their surroundings, providing the basis for map-based localization, obstacle avoidance, and loop closure. The dominant modalities in this category are LIDAR and RGB-D cameras, though acoustic sonars, stereo rigs, and event cameras also contribute in specific niches.

LIDAR (Light Detection and Ranging) sensors emit laser pulses and measure their return time to obtain dense range scans. Two-dimensional LIDAR has long been used with occupancy grid maps and scan-matching techniques for robust indoor localization (Thrun et al. 2005; Olson 2009). Three-dimensional spinning and solid-state LIDARs extend this principle to full 3D structure, supporting odometry and SLAM in complex outdoor environments (Zhang and Singh 2014; Shan and Englot 2018). Modern LIDAR odometry and mapping systems exploit geometric primitives such as planes and edges, combined with graph-based back-ends, to achieve high accuracy and robustness at moderate computational cost.

RGB-D cameras, such as the Microsoft Kinect and Intel RealSense, provide aligned colour and depth images at video rates. They enable dense or semi-dense SLAM in structured indoor spaces by fusing depth measurements into voxel grids, surfel maps, or truncated signed distance fields (Endres et al. 2012; Cadena et al. 2016). Compared with LIDAR, RGB-D sensors are compact and relatively inexpensive, but offer a limited range and field of view and are sensitive to ambient lighting and reflective surfaces.

Geometry sensors are tightly coupled to the map representations of Sect. 3. LIDAR and RGB-D measurements

create and update metric maps, while their derived features and surfaces provide constraints for graph-based estimators (Sect. 2.4). Their strengths include high spatial accuracy and the ability to support loop closure in both structured and unstructured environments. Limitations include cost and power consumption (for high-end LIDARs), sensitivity to environmental conditions, and the computational effort required for large-scale mapping.

4.4 Semantic perception and high-level cues for localization

Semantic perception does not constitute a separate physical sensing modality; instead, it denotes the set of processing modules that infer high-level information—such as object classes, room categories, or road semantics—from raw sensor signals. These modules operate on images, depth maps, point clouds, or other measurements to produce semantic labels, object instances, or relational structure that can support localization and mapping at a more abstract level.

A prominent example is SemanticFusion, which couples dense RGB-D SLAM with real-time semantic segmentation to build maps where each surface element is labelled with a semantic class (Mccormac et al. 2017). Variants extend this idea to object-centric and room-centric maps, enabling robots to reason about "being in the kitchen" or "near the table" rather than solely in metric coordinates (Bowman et al. 2017; Michael et al. 2022). Other work combines lidar or multi-modal data with deep networks to derive semantic point clouds, lane markings, or traversability estimates for autonomous driving and outdoor navigation (Jiang et al. 2023; Chen et al. 2021).

In the context of this survey, semantic perception serves two main roles. First, it enriches the map representations of Sect. 3.3 with labels and object instances, thereby enabling semantic map-based localization and topological reasoning. Second, it provides high-level cues to the learning-augmented estimators of Sect. 2.5, for example by suggesting loop-closure candidates based on object or scene similarity, or by conditioning motion priors on the current room type or task.

Table 6 Summary comparison of localization sensor categories. Qualitative levels (L, M, H) follow the conventions in Table 1.

Sensor category	Acc.	Rob.	Cost	Scale
Relative pose sensors (odometry, IMU, VO, VIO)	M	L–M	L–M	H
Global positioning sensors (GNSS, UWB)	M–H	M	M	H
Environmental geometry sensors (lidar, RGB-D)	H	M–H	M–H	M–H
Semantic perception modules (vision/lidar + learning)	M–H	M–H	M–H	M

The benefits of semantic perception include improved robustness to certain appearance changes (e.g., lighting, minor rearrangements), more human-interpretable localization outputs, and tighter coupling between localization and planning. At the same time, semantic modules depend on substantial training data, may not generalize well to unseen environments, and introduce additional computational load and potential failure modes when their predictions are incorrect. For these reasons, semantic perception is best viewed as an additional layer of information that complements, rather than replaces, geometric and relative sensing.

4.5 Summary and design implications

Table 6 summarizes the four sensor categories considered in this section in terms of accuracy, robustness, computational cost, and scalability, following the qualitative conventions introduced in Table 1. Table 7 complements this with a unified view of typical strengths and limitations, including how each category interacts with estimator families and map types.

Sensor configurations in typical scenarios. Typical localization deployments rely on specific combinations of the sensor categories reviewed above:

- *Indoor service robot in a structured building.* A common configuration combines wheel encoders and an IMU for high-rate relative pose, a 2D LIDAR or RGB-D camera for environmental geometry, and optionally UWB anchors for global anchoring in critical areas. Encoders and IMU feed a filtering or VIO front-end, while LIDAR/RGB-D support metric map-based localization and loop closure.
- *Autonomous vehicle in urban or highway environments.* Automotive platforms often fuse GNSS (with RTK or PPP when available), IMU, wheel odometry, multi-beam LIDAR, and cameras. GNSS provides global anchors when signals are reliable; IMU and odometry bridge short outages; LIDAR and cameras support detailed geometric and semantic mapping for lane-level localization and obstacle avoidance. Semantic perception modules extract lanes, traffic signs, and dynamic objects that constrain the estimator at a higher level.
- *GPS-denied subterranean, forest, or industrial sites.* In tunnels, mines, forests, or metal-rich industrial plants, GNSS is unavailable or unreliable. Robots rely on encoders, IMUs, LIDAR or RGB-D, and sometimes UWB anchors or Wi-Fi/Bluetooth ranging where infrastructure exists. Robust LIDAR/RGB-D-based SLAM and, when needed, particle-based global localization become central, while semantic perception can assist with loop

Table 7 Unified comparison of sensor categories in terms of typical strengths and limitations. Representative works are indicated in parentheses

Sensor category (representative works)	Typical strengths	Typical limitations
Relative pose sensors (odometry, IMU, VO, VIO) (Borenstein and Feng 1996; Borenstein 1995; Thrun et al. 2005; Titterton and Weston 2004; Groves 2013; Scaramuzza and Fraundorfer 2011; Mourikis and Roumeliotis 2007; Leutenegger et al. 2015; Qin et al. 2018)	High-rate motion estimates with low latency; inexpensive hardware for encoders and cameras; essential for control and for bridging gaps between sparser global or map-based updates; well understood integration in filtering and graph-based frameworks	Unbounded drift without external correction; sensitivity to slip, bias, illumination, and texture; limited robustness in highly dynamic or visually degraded environments; cannot provide absolute pose on their own
Global positioning sensors (GNSS, UWB) (Groves 2013; Ahmed et al. 2025; Alarifi et al. 2016; Al-Okby et al. 2024; Zhao et al. 2020)	Provide absolute pose anchors in a global frame; RTK/PPP GNSS and UWB can achieve decimetre- to centimetre-level accuracy; simplify large-scale mapping and multi-robot coordination; naturally complement relative and geometric sensors	Coverage gaps and degraded performance in urban canyons, indoors, or subterranean environments; dependence on satellite visibility or infrastructure deployment; susceptibility to multipath and interference; limited semantic information
Environmental geometry sensors (lidar, RGB-D) (Thrun et al. 2005; Olson 2009; Zhang and Singh 2014; Shan and Englot 2018; Endres et al. 2012; Cadena et al. 2016)	Provide accurate geometric structure for map-based localization and loop closure; support robust SLAM in structured and unstructured environments; directly feed the metric and hybrid maps of Sect. 3; widely used in safety-critical applications	High-end lidars incur significant cost, power, and data rates; performance depends on sufficient structural features and line-of-sight; RGB-D sensors have limited range and are sensitive to ambient lighting; large-scale mapping requires efficient data structures and computational resources
Semantic perception modules (vision/lidar + learning) (Mccormac et al. 2017; Bowman et al. 2017; Michael et al. 2022; Jiang et al. 2023; Chen et al. 2021)	Provide human-interpretable labels and high-level context (rooms, objects, road semantics); improve robustness under certain appearance changes; enable semantic maps and topological reasoning; connect localization to task-level planning and human instructions	Depend on large annotated datasets and may not generalize to novel environments; introduce additional latency and computational cost; predictions can be brittle under domain shift; require careful integration with geometric and probabilistic estimators to maintain consistency and safety

closure and environment understanding under poor geometric structure.

- *Semantics-heavy domestic or social robotics.* Service robots in homes, hospitals, or offices benefit from cameras and depth sensors coupled with strong semantic perception modules. Relative pose sensing (VO/VIO) and occasional LIDAR or RGB-D mapping provide geometric consistency, while semantic segmentation and object recognition feed semantic maps and support language-guided navigation and interaction with humans.

From a design perspective, relative sensors are indispensable for any platform that must track motion at high rates, but they should be treated as short-term priors rather than sources of globally consistent pose. Global positioning sensors are highly attractive whenever satellite coverage or infrastructure can be relied upon, as they greatly simplify the estimator design by providing absolute anchors that constrain drift. Environmental geometry sensors are the workhorse for map-based localization, particularly in safety-critical or GPS-denied scenarios, while semantic perception modules enrich these maps with context and support interaction with humans and high-level tasks.

In many practical systems, no single modality is sufficient. Instead, designers combine relative, global, geometric, and semantic cues within the estimator families of Sect. 2 and the multi-layer maps of Sect. 3. Achieving principled fusion across these levels—including consistent uncertainty handling, robust outlier rejection, and efficient use of learned semantic information—remains a central challenge and a key opportunity for future localization research, as further discussed in Sect. 5.

5 Trends, challenges, and future directions

The preceding sections reviewed core estimation techniques (Sect. 2), map representations (Sect. 3), and sensor modalities (Sect. 4) through a functional lens. Taken together, they show that modern localization is no longer a single algorithmic choice, but a system-level design problem that couples estimators, multi-layer maps, and heterogeneous sensors.

This section synthesizes cross-cutting trends that are reshaping the field and identifies open challenges that remain even in state-of-the-art systems. Rather than attempting an exhaustive list, we focus on directions that

are directly suggested by the taxonomies developed earlier: learning-augmented estimation, implicit and multi-layer maps, cross-modal and multi-robot localization, resource-aware deployment, and the integration of semantics into localization pipelines.

5.1 Emerging research trends

The landscape of robot localization is being reshaped by advances in machine learning, representation learning, and distributed systems, as well as by the increasing deployment of robots in complex, human-centric environments. Several trends are particularly prominent.

Learning-augmented estimation and active localization. Section 2.5 highlighted how learning is increasingly used to enhance, rather than replace, classical estimators. Learned measurement models, priors, and semantic cues are now routinely integrated into filters and graph-based back-ends, improving robustness to appearance change and complex sensor characteristics. A related trend is *active localization*, in which planning and estimation are co-designed: the robot selects actions that are informative for localization, for example by trading off exploration and exploitation or by explicitly planning to disambiguate perceptual aliasing. Combining learning-augmented estimators with principled active policies remains a central opportunity, especially in environments where observations are expensive or safety-critical.

Implicit and multi-layer map representations. As reviewed in Sects. 3.3 and 3.4, there is growing interest in maps that go beyond a single metric grid or point cloud. Multi-layer maps combine local metric submaps, topological place graphs, and semantic or object-centric layers, enabling systems to operate at multiple levels of abstraction. In parallel, implicit neural representations and neural fields offer compact, continuous encodings of geometry and appearance. A major trend is the integration of these structures: classical metric and topological maps provide strong geometric priors and reliability, while learned and implicit layers capture fine-scale details, semantics, and long-range correlations. Designing estimators that can consistently reason across such layered maps is an active research frontier.

Cross-modal, cross-embodiment, and multi-robot localization. Section 4 emphasized that no single sensor family is sufficient across all regimes. Recent work therefore focuses on cross-modal localization, where systems exploit overlapping structure between different sensor suites (e.g., LIDAR-only and camera-only platforms) and transfer localization capabilities across robots or datasets. At the same time, multi-robot SLAM and distributed localization architectures aim to share information efficiently between agents while respecting bandwidth and privacy constraints. These trends point toward localization systems that can adapt to heterogeneous platforms and dynamically changing sensor configurations without extensive manual retuning.

Resource-aware and safety-critical localization. As localization moves from research prototypes to deployed products, constraints on computation, energy, and safety become central. Resource-aware localization seeks to maintain reliable performance on embedded platforms by adapting estimator complexity, map resolution, and sensing rates to available resources. Neuromorphic and event-based sensing, hardware acceleration, and approximate inference methods are being explored to reduce latency and power consumption. At the same time, safety-critical domains such as autonomous driving, aerial inspection, and collaborative manipulation demand rigorous treatment of uncertainty, fail-safe behaviours under localization loss, and interfaces to formal verification and certification processes.

Semantics, language, and human-centric localization. Finally, there is a growing interest in localization systems that can reason not only about coordinates, but also about human-interpretable concepts such as rooms, objects, and tasks. Semantic and object-centric maps (Sect. 3.3) provide a substrate for this, while VLMs and LLMs offer mechanisms for grounding natural-language descriptions and instructions in the robot's spatial representation. Early work explores using such models for semantic loop closure, scene description, and language-guided navigation, but robustly integrating high-level, symbolic reasoning with low-level geometric estimation remains an open and challenging direction.

5.2 Open challenges and future directions

Despite substantial progress, several fundamental challenges remain. We group them here into five themes that cut across estimators, maps, and sensing.

Robustness in extreme and dynamic environments. Many localization systems still struggle in the presence of severe dynamics (moving crowds, traffic, foliage), adverse weather and lighting, or structural changes. While robust cost functions, outlier rejection, and semantic filtering help, achieving sustained performance under such conditions remains difficult. A key open question is how to formally characterize and guarantee robustness when both the

environment and the sensor suite may deviate significantly from the conditions seen during development or training.

Scalability and computational efficiency. Large-scale, long-term deployments generate vast amounts of data and require estimators and map structures that can grow gracefully without sacrificing real-time performance. Sparse graph-based formulations, submapping, and hierarchical maps address part of this challenge, but trade-offs between map resolution, memory, and estimator complexity are still poorly understood. There is a need for principled strategies to adaptively select what to store, what to marginalize, and what to discard, while preserving the information most relevant for future localization and planning.

Data efficiency, generalization, and evaluation. Learning-augmented estimators and semantic maps require substantial training data, which may not be available for every environment or sensing configuration. Methods that can transfer across domains, leverage self-supervision, or exploit simulation-to-real pipelines are promising but not yet mature. In parallel, the community lacks standardized, cross-modal benchmarks that systematically probe failure modes, long-term drift, and recovery from localization loss across different estimator, map, and sensor combinations. Developing such benchmarks and evaluation protocols is essential for comparing methods and for supporting deployment in safety-critical settings.

Uncertainty quantification, verification, and interpretability. Classical probabilistic estimators provide explicit uncertainty representations, but these are often violated in practice due to modelling approximations, linearization, and unmodelled correlations. Learning-based components further complicate the picture, as their uncertainty estimates are frequently miscalibrated. There is an open need for methods that can provide reliable uncertainty quantification for hybrid systems combining learned and model-based components, and for tools that allow practitioners to diagnose, verify, and explain localization behaviour to both engineers and non-expert stakeholders.

Ethical, privacy, and societal considerations. As robots increasingly operate in homes, workplaces, and public spaces, localization and mapping acquire explicit ethical and legal dimensions. High-resolution geometric and semantic maps can inadvertently capture and store sensitive information about people, their possessions, and their routines, raising questions about data minimization, access control, retention policies, and the potential misuse of detailed spatial data. Techniques such as on-device processing, selective obfuscation or abstraction of sensitive regions, and privacy-preserving map sharing are beginning to appear, but remain far from standard practice. Close collaboration with legal and ethics communities will be required to translate emerging regulatory frameworks into concrete design guidelines and technical mechanisms for privacy-aware localization.

Persistent localization and mapping in domestic, clinical, and workplace environments thus raise privacy concerns that go beyond classical safety and reliability. Survey-based studies on social robots show that users' willingness to adopt such systems depends strongly on how invasively the platform collects and processes personal data: privacy-friendly configurations lead to significantly higher use intentions than privacy-invasive ones (Lutz & Tamò-Larrieux 2021). Concrete incidents underline these concerns. For example, a widely discussed case involving development versions of a consumer robot vacuum demonstrated how images captured for mapping and obstacle avoidance, including highly intimate scenes, entered cloud-based annotation workflows and ultimately circulated on social media platforms (Guo 2022). Such episodes illustrate that even ostensibly benign localization and mapping functions can expose sensitive information about a person's home layout, possessions, and everyday activities.

These risks have motivated a growing line of work on privacy-preserving localization and SLAM. For camera-based systems, several approaches aim to retain geometric utility while obfuscating appearance. Line-cloud map representations replace dense 3D point clouds with 3D line clouds to reduce the amount of recoverable scene detail while still supporting accurate six-degree-of-freedom pose estimation (Speciale et al. 2019). At the feature level, adversarial affine subspace embeddings provide privacy-preserving local descriptors that remain useful for visual localization and mapping but substantially hinder reconstruction of the original images (Dusmanu et al. 2021). More recently, dual-component visual SLAM pipelines have combined object-level redaction with encrypted line-cloud maps to support

Table 8 Key future directions in robot localization, with indicative benefits and challenges

Research direction	Main benefits	Key challenges
Cross-modal and adaptive localization	Robust operation under changing sensors and environments; graceful degradation when individual modalities fail; reuse of maps and models across platforms	Modelling heterogeneous sensors and maps in a unified framework; handling out-of-distribution behaviour; designing benchmarks that reflect realistic cross-modal deployment
Learning-augmented and semantic localization	Richer perception and context-awareness; improved robustness to appearance changes; more natural interaction through semantic maps and language grounding	Data and compute requirements; generalization beyond training domains; integrating learned components with probabilistic estimators while preserving consistency and safety
Scalable, long-term, and distributed localization	Operation over large spatial and temporal scales; support for multi-robot systems and shared infrastructure; efficient reuse of prior maps and experience	Managing memory and communication budgets; maintaining consistency under partial, delayed, or conflicting information; robust map merging and versioning
Uncertainty-aware and safety-critical localization	Better calibrated decision-making in safety-critical tasks; explicit reasoning about failure modes and fall-back strategies; interfaces to verification and certification	Developing tractable yet faithful uncertainty models for hybrid systems; validating uncertainty estimates in practice; integrating localization guarantees into system-level safety cases
Ethics- and privacy-aware mapping and localization	Increased trust and acceptability in human-centric environments; alignment with regulatory requirements; reduced risk of misuse of detailed maps	Balancing privacy with the need for detailed context; formalizing privacy requirements in technical terms; deploying privacy-preserving mechanisms without degrading localization performance excessively

edge-assisted localization under explicit privacy constraints (Yang et al. 2025). While these methods have so far been developed mainly for indoor and urban scenarios, they indicate a technical toolbox that could be integrated into future localization pipelines whenever high-resolution mapping interacts with identifiable individuals or sensitive assets Table 8.

5.3 Future outlook and vision

Looking ahead, robot localization is likely to evolve from a predominantly geometric estimation problem into a broader capability that combines geometry, semantics, and high-level reasoning. The taxonomies in Sects. 2–4 suggest that future systems will increasingly rely on multi-layer maps, hybrid estimators that blend learning with probabilistic inference, and sensor suites that are adapted to specific operating regimes but able to interoperate through shared abstractions.

A central step toward this vision is tighter coupling between localization, mapping, and task-level decision-making. Estimators will be expected not only to provide accurate poses, but also to expose uncertainty, semantic structure, and failure indicators in a form that planners and supervisory modules can use. Advances in large language models (LLMs) and vision-language models (VLMs) provide additional tools for connecting human instructions, textual knowledge, and spatial representations, but their integration must be guided by careful consideration of reliability, safety, and privacy.

Ultimately, progress in robot localization will be measured not only by reductions in average pose error on benchmark datasets, but by the ability of systems to operate robustly, transparently, and responsibly in the diverse environments where robots are deployed. The research directions outlined above aim to move the field toward such systems, in which localization is a reliable, interpretable component of autonomous behaviour rather than a brittle, opaque subsystem.

Appendix A Abbreviations

In addition, all technical abbreviations are defined at their first occurrence in the text to ensure clarity and uniformity of terminology, all abbreviations used throughout this survey, presented in Table 9

Table 9 List of Abbreviations

Abbreviation	Full Term	Abbreviation	Full Term
Acoustic	Acoustic (Sensor Type)	INS	Inertial Navigation System
AGV	Automated Guided Vehicle	IoT	Internet of Things
AI	Artificial Intelligence	KLD-sampling	Kullback-Leibler Divergence Sampling
ANNs	Artificial Neural Networks	Landmark	Landmark Map Type
ARTMAP	Adaptive Resonance Theory Map	Laser	Laser (Sensor Modality)
AS	Ant Systems	LBL	Long Baseline
Beac.	Beacons	LiDAR	Light Detection and Ranging
BLE	Bluetooth Low Energy	LLMs	Large Language Models
BoW	Bag-of-Words	LMBP	Levenberg-Marquardt Back-Propagation
BPANN	Back-Propagation Artificial Neural Network	LTE	Long Term Evolution
Cam.	Camera	Mag.	Magnetic (Sensor Modality)
CI	Covariance Intersection	MCL	Monte Carlo Localization
CNN	Convolutional Neural Network	MHE	Moving Horizon Estimation
Colour-Depth	Colour-Depth Cameras	Multi	Multiple
CSI	Channel State Information	NARF	Normal Alignment Radial Feature
DBSCAN	Density-Based Spatial Clustering of Applications with Noise	NDT	Normal Distributions Transform
DoA	Direction of Arrival	NSGA-II	Non-dominated Sorting Genetic Algorithm II
DP	Dynamic Programming	Odom.	Odometry
DR	Dead Reckoning	PCA	Principal Component Analysis
DVL	Doppler Velocity Log	PF	Particle Filter
EAs	Evolutionary Algorithms	PFH	Point Feature Histograms
EFIR	Extended Finite Impulse Response	Piezo.	Piezoelectric Sensor (Flow Sensor)
EKF	Extended Kalman Filter	POF	Particle Swarm Optimization Filter
ELLMER	Embodied LLM for Multi-step Execution and Retrieval	PSO	Particle Swarm Optimization
EM	Expectation-Maximization	RADAR	Radio Detection and Ranging
Enc.	Wheel Encoders	RAMIS	Robot-Assisted Minimally Invasive Surgery
ENN	Evolutionary Neural Networks	RANSAC	Random Sample Consensus
Extero.	Exteroceptive	Range	Range (Sensor Modality)
Feature-based	Feature-based Map/Method Type	RBPFs	Rao-Blackwellized Particle Filters
FIS	Fuzzy Inference System	RF	Radio Frequency
FLAKF	Fuzzy Logic Adaptive Kalman Filter	RFID	Radio-Frequency Identification
GAs	Genetic Algorithms	RGB-D	Red/Green/Blue + Depth (Camera/Data)
GGM	Generalized Gaussian Mixture	RSSI	Received Signal Strength Indicator
GIS	Geographic Information System	SBL	Short Baseline
GNSS	Global Navigation Satellite Systems	Sem.	Semantic (Map Type)
GPS	Global Positioning System	SeqSLAM	Sequential SLAM (Algorithm)
GVG	Generalized Voronoi Graphs	SeqSLAM++	Enhanced SeqSLAM (Algorithm)
HAYAI	High-speed and Yet Accurate Indoor/Outdoor Tracking	SF	Sensor Fusion
HMM	Hidden Markov Model	SFVs	Statistical Feature Vectors
HOG	Histograms of Oriented Gradients	SIFT	Scale-Invariant Feature Transform
ICP	Iterative Closest Points	SIVIA	Set Inversion Via Interval Analysis
IGA	Island Model Genetic Algorithm	SLAM	Simultaneous Localization and Mapping
iGS	Indoor GPS system (using RFID/ultrasound)	Sonar	Sonar (Sensor Modality)
IMU	Inertial Measurement Unit	VLMs	Vision-Language Models
Infra.	Infrastructure-dependent (Map Type/Sensors)		

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References

Aggarwal, P., Syed, Z., Niu, X., El-Sheimy, N.: A standard testing and calibration procedure for low cost MEMS inertial sensors and units. J. Navig. **61**(2), 323–336 (2008). https://doi.org/10.1017/S002232980800508X

Ahmed, W.A., Isiaka, L.A., Mala, B., Olatoyinbo, S.F.: Evolution of GNSS/GPS technology and its applications from ancient times to the present. Next Res. **2**(3), 100387 (2025). https://doi.org/10.1016/j.nexres.2025.100387

Alarifi, A., Al-Salman, A., Alsaleh, M., Alnafessah, A., Al-Hadhrami, A., Al-Ammar, M., Al-Khalifa, H.: Ultra-Wideband (UWB) Indoor Positioning System: Current status and future perspective. J. Commun. **11**(12), 1092–1105 (2016). https://doi.org/10.12720/jcm.11.12.1092-1105

Al-Okby, M., Junginger, S., Roddelkopf, T., Thurow, K.: UWB-based real-time Indoor Positioning Systems: A Comprehensive Review. Applied Sciences **14**(23), 11005 (2024). https://doi.org/10.3390/app142311005

Arulampalam, M.S., Maskell, S., Gordon, N., Clapp, T.: A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking. IEEE Trans. Signal Process. **50**(2), 174–188 (2002). https://doi.org/10.1109/78.978374

Bader, K., Lussier, B., Schön, W.: A fault tolerant architecture for data fusion: A real application of Kalman filters for mobile robot localization. Robot. Auton. Syst. **88**, 11–23 (2017). https://doi.org/10.1016/j.robot.2016.11.015

Bailey, T., Durrant-Whyte, H.: Simultaneous localization and mapping (SLAM): part II. IEEE Robotics & Automation Magazine **13**(3), 108–117 (2006). https://doi.org/10.1109/MRA.2006.1678144

Besl, P.J., McKay, N.D.: A method for registration of 3-d shapes. IEEE Trans. Pattern Anal. Mach. Intell. **14**(2), 239–256 (1992). https://doi.org/10.1109/34.121791

Biber, P., Strasser, W.: The normal distributions transform: A new approach to laser scan matching. In: Proc. IEEE/RSJ Int. Conf. Intell. Rob. Syst. (IROS), pp. 2743–2748 (2003). https://doi.org/10.1109/IROS.2003.1249258

Borenstein, J.: Intemal correction of dead-reckoning errors with a dual-drive compliant linkage mobil robot. J. Robot. Syst. **12**(4), 257–273 (1995). https://doi.org/10.1002/rob.4620120405

Borenstein, J., Feng, L.: Measurement and correction of systematic odometry errors in mobile robots. IEEE Trans. Robot. Autom. **12**(6), 869–880 (1996). https://doi.org/10.1109/70.544770

Borges, G.A., Aldon, M.-J.: Robustified estimation algorithms for mobile robot localization based on geometrical environment maps. Robotics and Autonomous Systems **45**(3), 131–159 (2003). https://doi.org/10.1016/j.robot.2003.09.003

Bowman, S.L., Paul, R., Dellaert, F., Roy, N.: Probabilistic data association for semantic slam. In: IEEE Int. Conf. Rob. Aut. (ICRA), pp. 1722–1729 (2017). https://doi.org/10.1109/ICRA.2017.7989205

Cadena, C., Carlone, L., Carrillo, H., Latif, Y., Scaramuzza, D., Neira, J., Reid, I., Leonard, J.J.: Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age. IEEE Transactions on Robotics **32**(6), 1309–1332 (2016). https://doi.org/10.1109/TRO.2016.2624754

Campos, C., Elvira, R., Rodriguez, J.J.G., Montiel, J.M., Tardos, J.: Orb-slam3: An accurate open-source library for visual, visual–inertial, and multimap slam. IEEE Transactions on Robotics **37**(6), 1874–1890 (2021). https://doi.org/10.1109/TRO.2021.3075644

Carrillo-Arce, L.C., Nerurkar, E.D., Gordillo, J.L., Roumeliotis, S.I.: Decentralized multi-robot cooperative localization using covariance intersection. In: IEEE/RSJ Int. Conf. Intell. Rob. Syst. pp. 1412–1417 (2013). https://doi.org/10.1109/IROS.2013.6696534

Chang, T.-K., Chen, K., Mehta, A.: Resilient and Consistent Multirobot Cooperative Localization With Covariance Intersection. IEEE Trans. Rob. **38**(1), 197–208 (2021)

Chen, G., Wang, B., Wang, X., Deng, H., Wang, B., Zhang, S.: PSF-LO: Parameterized Semantic Features Based Lidar Odometry (2021). arXiv:2010.13355

Chen, C., Wang, B., Lu, C.X., Trigoni, N., Markham, A.: Deep learning for visual localization and mapping: A survey. IEEE Trans. Neural Netw. Learn. Syst. (2023). https://doi.org/10.1109/TNNLS.2023.3309809

Ching-Chih, T., Chun-Chieh, C., Feng-Chun, T.: Cooperative Localization Using Fuzzy Decentralized Extended Information Filtering for Homogenous Omnidirectional Mobile Multi-robot. System (2015). https://doi.org/10.3233/978-1-61499-522-7-343

Choi, Y., Di Marco, P., Park, P.: Communication-Aware Graph Neural Network for Multi-Agent Reinforcement Learning. IEEE Access (2025). https://doi.org/10.1109/ACCESS.2025.3554736

Chong, K.S., Kleeman, L.: Accurate odometry and error modelling for a mobile robot. In: Proc. IEEE Int. Conf. Rob. Auto. pp. 2783–2788 (1997). https://doi.org/10.1109/ROBOT.1997.606708

Cox, I.J.: In: Cox, I.J., Wilfong, G.T. (eds.) Blanche: Position Estimation for an Autonomous Robot Vehicle, pp. 221–228. Springer, New York, NY (1990). https://doi.org/10.1007/978-1-4613-8997-2_17

Cummins, M., Newman, P.: Fab-map: Probabilistic localization and mapping in the space of appearance. The International Journal of Robotics Research **27**, 647–665 (2008). https://doi.org/10.1177/0278364908090961

Dellaert, F.: Factor graphs and gtsam: A hands-on introduction. Technical Report GT-RIM-CP_R-2012-002, Georgia Institute of Technology (2012)

Dellaert, F., Kaess, M.: Square root sam: Simultaneous localization and mapping via square root information smoothing. In: Proc. Rob.: Sci. Syst. (RSS), pp. 177–184 (2006). https://doi.org/10.15607/RSS.2005.I.024

Dellaert, F., Fox, D., Burgard, W., Thrun, S.: Monte Carlo Localization for Mobile Robots. Proceedings of the 1999 IEEE International Conference on Robotics and Automation (ICRA), 1322–1328 (1999) https://doi.org/10.1109/ROBOT.1999.772516

Doucet, A., Freitas, N., Gordon, N.: Sequential Monte Carlo Methods in Practice. Springer, New York (2001)

Dourado, C.M.J.M., da Silva, S.P.P., da Nóbrega, R.V.M., Barros, A.C.S., Sangaiah, A.K., Rebouças Filho, P.P., de Albuquerque, V.H.C.: A new approach for mobile robot localization based on an online iot system. Future Generation Computer Systems **100**, 859–881 (2019). https://doi.org/10.1016/j.future.2019.05.074

Dusmanu, M., Schonberger, J.L., Sinha, S.N., Pollefeys, M.: Privacy-preserving Image Features via Adversarial Affine Subspace Embeddings. In: Proc. IEEE/CVF Conf. Comput. Vision and Pattern Recog. (CVPR), pp. 14267–14277 (2021). https://doi.org/10.1109/CVPR46437.2021.01404

Endres, F., Hess, J., Engelhard, N., Sturm, J., Cremers, D., Burgard, W.: An evaluation of the rgb-d slam system, 1691–1696 (2012) https://doi.org/10.1109/ICRA.2012.6225199

Engel, J., Schöps, T., Cremers, D.: LSD-SLAM: Large-scale direct Monocular SLAM. In: Proc. European Conf. Comput. Vision (ECCV), pp. 834–849. Springer, ??? (2014). https://doi.org/10.1007/978-3-319-10590-1_54

Erfani, S., Jafari, A., Hajiahmad, A.: Comparison of two data fusion methods for localization of wheeled mobile robot in farm conditions. Artif. Intell. Agric. **1**, 48–55 (2019). https://doi.org/10.1016/j.aiia.2019.05.002

Fox, D., Burgard, W., Thrun, S.: Active Markov localization for mobile robots. Robotics and Autonomous Systems **25**(3), 195–207 (1998). https://doi.org/10.1016/S0921-8890(98)00049-9

Fuentes-Pacheco, J., Ruiz-Ascencio, J., Rendón-Mancha, J.M.: Visual simultaneous localization and mapping: A survey. Artif. Intell. Rev. **43**(1), 55–81 (2015). https://doi.org/10.1007/s10462-012-9365-8

Galvez-López, D., Tardós, J.D.: Bags of binary words for fast place recognition in image sequences. IEEE Transactions on Robotics **28**, 1188–1197 (2012). https://doi.org/10.1109/TRO.2012.2197158

Gellert, K., Schlögl, E.: Parameter Learning and Change Detection Using a Particle Filter With Accelerated Adaptation (2018). arXiv:1806.05387

Gordon, N.J., Salmond, D.J., Smith, A.F.M.: Novel approach to nonlinear/non-Gaussian Bayesian state estimation. IEE Proceedings F Radar and Signal Processing **140**, 107–113 (1993). https://doi.org/10.1049/ip-f-2.1993.0015

Grisetti, G., Stachniss, C., Burgard, W.: Improved techniques for grid mapping with rao-blackwellized particle filters. IEEE Trans. Rob. **23**(1), 34–46 (2007). https://doi.org/10.1109/TRO.2006.889486

Grisetti, G., Kümmerle, R., Stachniss, C., Burgard, W.: A tutorial on graph-based slam. IEEE Intell. Transp. Syst. Magaz. **2**, 31–43 (2010). https://doi.org/10.1109/MITS.2010.939925

Groves, P.: Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems, Second Edition, (2013)

Guivant, J.E., Masson, F.R., Nebot, E.M.: Simultaneous localization and map building using natural features and absolute information. Robotics and Autonomous Systems **40**(2), 79–90 (2002). https://doi.org/10.1016/S0921-8890(02)00233-6

Guo, E.: A Roomba Recorded a Woman on the Toilet. How Did Screenshots End Up on Facebook? (2022). https://www.technologyreview.com/2022/12/19/1065306/roomba-irobot-robot-vacuums-artificial-intelligence-training-data-privacy/

Hashimoto, M., Oba, F., Tomiie, T.: Mobile robot localization using color signboard. Mechatronics **9**(6), 633–656 (1999). https://doi.org/10.1016/S0957-4158(99)00012-4

Hess, W., Kohler, D., Rapp, H., Andor, D.: Real-time loop closure in 2d lidar slam. In: Proceedings of the IEEE Int. Conf. Rob. Auto. (ICRA), pp. 1271–1278 (2016). https://doi.org/10.1109/ICRA.2016.7487258

Hornung, A., Wurm, K.M., Bennewitz, M., Stachniss, C., Burgard, W.: Octomap: An efficient probabilistic 3d mapping framework based on octrees. Auton. Rob. **34**, 189–206 (2013). https://doi.org/10.1007/s10514-012-9321-0

Howard, A., Mataric, M.J., Sukhatme, G.S.: Relaxation on a mesh: A formalism for generalized localization. In: Proc. IEEE/RSJ Int. Conf. Intell. Rob. Syst. (IROS), pp. 1055–1060 (2001). https://doi.org/10.1109/IROS.2001.976306

Jiang, K., Gao, S., Zhang, X., Li, J., Guo, Y., Liu, S., Li, C., Wang, J.: SELVO: A semantic-enhanced Lidar-Visual Odometry. In: 2023 IEEE/RSJ Int. Conf. Intell. Rob. Syst. (IROS), pp. 1473–1480 (2023). https://doi.org/10.1109/IROS55552.2023.10341419

Julier, S.J., Uhlmann, J.K.: Unscented filtering and nonlinear estimation. Proc. IEEE **92**(3), 401–422 (2004). https://doi.org/10.1109/JPROC.2003.823141

Kaess, M., Ranganathan, A., Dellaert, F.: isam: Incremental smoothing and mapping. IEEE Trans. Rob. **24**(6), 1365–1378 (2008). https://doi.org/10.1109/TRO.2008.2006706

Kaess, M., Johannsson, H., Roberts, R., Ila, V., Leonard, J.J., Dellaert, F.: isam2: Incremental smoothing and mapping using the bayes tree. Int. J. Rob. Res. **31**(2), 216–235 (2012). https://doi.org/10.1177/0278364911430419

Kalman, R.E.: A new approach to linear filtering and prediction problems. Journal of Basic Engineering **82**(1), 35–45 (1960)

Kelly, A.: General solution for linearized systematic error propagation in vehicle odometry. In: Proc. IEEE/RSJ Int. Conf. Intell. Rob. Syst. pp. 1938–1945 (2001). https://doi.org/10.1109/IROS.2001.976357

Kendall, A., Grimes, M., Cipolla, R.: PoseNet: A Convolutional Network for Real-Time 6-DOF Camera Relocalization (2016). arXiv:1505.07427

Koenig, S., Simmons, R.: Unsupervised Learning of Probabilistic Models for Robot Navigation. In: Proc. IEEE Int. Conf. Rob. Auto. (ICRA), pp. 2301–2308 (1996).https://doi.org/10.1109/ROBOT.1996.506901

Lee, H., Jung, J., Choi, K., Park, J., Myung, H.: Fuzzy-logic-assisted interacting multiple model (FLAIMM) for mobile robot localization. Robot. Auton. Syst. **60**(12), 1592–1606 (2012). https://doi.org/10.1016/j.robot.2012.09.018

Lenser, S., Veloso, M.: Sensor resetting localization for poorly modelled mobile robots. In: Proceedings 2000 ICRA. Millennium Conference. IEEE Int. Conf. Rob. Auto. Sympos. Proc (Cat. No.00CH37065), 2:1225–12322 (2000). https://doi.org/10.1109/ROBOT.2000.844766

Leutenegger, S., Lynen, S., Bosse, M., Siegwart, R., Furgale, P.: Keyframe-based visual–inertial odometry using nonlinear optimization. Int. J. Rob. Res. **34**(3), 314–334 (2015). https://doi.org/10.1177/0278364914554813

Lu, F., Milios, E.: Robot pose estimation in unknown environments by matching 2d range scans. Journal of Intelligent and Robotic Systems **18**(3), 249–275 (1997)

Lutz, C., Tamò-Larrieux, A.: Do privacy concerns about social robots affect use intentions? evidence from an experimental Vignette study. Frontiers in Robotics and AI **8**, 627958 (2021). https://doi.org/10.3389/frobt.2021.627958

Magnusson, M.: The three-dimensional normal distributions transform—an efficient representation for registration, surface analysis, and loop detection. PhD thesis, Örebro University (2009). http://urn.kb.se/resolve?urn=urn:nbn:se:oru:diva-4549

Martinelli, A.: The odometry error of a mobile robot with a synchronous drive system. IEEE Transactions on Robotics and Automation **18**(3), 399–405 (2002). https://doi.org/10.1109/tra.2002.1019477

Matsuo, K., Aoki, Y., Tsubouchi, T.: Keyframe-based visual localization using image descriptor matching. In: IEEE/SICE Int. Sympos. Syst. Integration, pp. 109–114 (2014)
Maybeck, P.S.: Stochastic Models, Estimation, and Control, vol. 1. Academic Press, New York (1979)
McCormac, J., Handa, A., Leutenegger, S., Davison, A.J.: Semanticfusion: Dense 3d semantic mapping with convolutional neural networks. In: IEEE Int. Conf. Rob. Auto. (ICRA), pp. 4628–4635 (2017).https://doi.org/10.1109/ICRA.2017.7989516
Mescheder, L., Oechsle, M., Niemeyer, M., Nowozin, S., Geiger, A.: Occupancy networks: Learning 3D reconstruction in Function Space. In: Proc. IEEE/CVF Conf. Comput. Vision and Pattern Recog. (CVPR), pp. 4460–4470 (2019).https://doi.org/10.1109/CVPR.2019.00459
Michael, E., Summers, T., Wood, T.A., Manzie, C., Shames, I.: Probabilistic data association for semantic slam at scale (2022) arXiv:2202.12802 [cs.RO]
Mildenhall, B., Srinivasan, P.P., Tancik, M., Barron, J.T., Ramamoorthi, R., Ng, R.: NeRF: Representing scenes as neural radiance fields for view synthesis. In: Proc. Eur. Conf. Comput. Vision (ECCV), pp. 405–421. Springer, ??? (2020). https://doi.org/10.1007/978-3-030-58452-8_24
Mohamed, S.A.S., Haghbayan, M.-H., Westerlund, T., Heikkonen, J., Tenhunen, H., Plosila, J.: A survey on odometry for autonomous navigation systems. IEEE Access **7**, 97466–97486 (2019). https://doi.org/10.1109/access.2019.2929133
Mourikis, A.I., Roumeliotis, S.I.: A Multi-State Constraint Kalman Filter for Vision-aided Inertial Navigation. In: Proc. IEEE Int. Conf. Rob. Auto. (ICRA), pp. 3565–3572 (2007)
Mun, J.-H., Jeon, M., Lee, B.-G.: Unsupervised learning for depth, ego-motion, and optical flow estimation using coupled consistency conditions. Sensors (2019). https://doi.org/10.3390/s19112459
Mur-Artal, R., Tardos, J.D.: ORB-SLAM2: An open-source SLAM system for monocular, stereo, and RGB-D cameras. IEEE Trans. Rob. **33**(5), 1255–1262 (2017). https://doi.org/10.1109/TRO.2017.2705109
Newcombe, R.A., Lovegrove, S., Davison, A.J.: DTAM: Dense tracking and mapping in real-time. In: Proc. IEEE Int. Conf. Comput. Vision (ICCV), pp. 2320–2327 (2011). https://doi.org/10.1109/ICCV.2011.6126513
Olson, E.: Real-time correlative scan matching. In: Proc. IEEE Int. Conf. Rob. Auto. (ICRA), pp. 4387–4393 (2009). https://doi.org/10.1109/ROBOT.2009.5152815
Olson, E., Leonard, J., Teller, S.: Fast iterative alignment of pose graphs with poor initial estimates, pp. 2262–2269 (2006). https://doi.org/10.1109/ROBOT.2006.1642040
Qin, T., Li, P., Shen, S.: VINS-Mono: A robust and versatile monocular visual-inertial state estimator. IEEE Trans. Rob **34**(4), 1004–1020 (2018). https://doi.org/10.1109/TRO.2018.2853739
Ristic, B., Arulampalam, S., Gordon, N.: Beyond the Kalman Filter: Particle Filters for Tracking Applications. Artech House, Norwood, MA (2004)
Rosinol, A., Abate, M., Chang, Y., Carlone, L.: Kimera: an open-source library for real-time metric-semantic localization and mapping. (2020). arXiv:1910.02490
Sarlin, P.-E., DeTone, D., Malisiewicz, T., Rabinovich, A.: SuperGlue: Learning Feature Matching with Graph Neural Networks (2020). arXiv:1911.11763
Scaramuzza, D., Fraundorfer, F.: Visual odometry [tutorial]. IEEE Robotics & Automation Magazine **18**(4), 80–92 (2011). https://doi.org/10.1109/MRA.2011.943233
Scaramuzza, D., Martinelli, A., Siegwart, R.: A flexible omnidirectional camera system for localization and mapping. IEEE Int. Conf. Intell. Rob. Syst. 5271–5276 (2006)
Segal, A., Hähnel, D., Thrun, S.: Generalized-icp. In: Proceedings of Robotics: Science and Systems (RSS) (2009). https://doi.org/10.15607/RSS.2009.V.021
Shan, T., Englot, B.: Lego-loam: Lightweight and ground-optimized lidar odometry and mapping on variable terrain, 4758–4765 (2018). https://doi.org/10.1109/IROS.2018.8594299
Shin, E.H.: Estimation techniques for low-cost inertial navigation. M.sc. thesis, Department of Geomatics Engineering, University of Calgary, Calgary, AB, Canada (2005)
Simon, D.: Optimal State Estimation: Kalman, H Infinity, and Nonlinear Approaches. John Wiley & Sons, Hoboken, NJ (2006)
Speciale, P., Schonberger, J.L., Kang, S.B., Sinha, S.N., Pollefeys, M.: Privacy preserving Image-Based Localization. In: Proc. IEEE/CVF Conf. Comput. Vision and Pattern Recog.(CVPR), pp. 5488–5498 (2019). https://doi.org/10.1109/CVPR.2019.00564
Suzuki, T.: Mobile robot localization with GNSS multipath detection using pseudorange residuals. Adv. Rob. **33**(12), 602–613 (2019). https://doi.org/10.1080/01691864.2019.1619622
Teslić, L., Škrjanc, I., Klančar, G.: Using a LRF sensor in the Kalman-filtering-based localization of a mobile robot. ISA Trans **49**(1), 145–153 (2010). https://doi.org/10.1016/j.isatra.2009.09.009
Thrun, S.: Learning metric-topological maps for indoor mobile robot navigation. Artif. Intell. **99**(1), 21–71 (1998)
Thrun, S., Fox, D., Burgard, W., Dellaert, F.: Robust Monte Carlo localization for mobile robots. Artif. Intell. **128**(1), 99–141 (2001). https://doi.org/10.1016/S0004-3702(01)00069-8
Thrun, S., Burgard, W., Fox, D.: Probabilistic Robotics. MIT Press, Cambridge, Mass (2005)
Titterton, D., Weston, J.: Strapdown Inertial Navigation Technology, 2nd edn The Institution of Engineering and Technology, London, UK; Stevenage, Herts, UK (2004). https://doi.org/10.1049/PBRA017E
Walch, F., Hazirbas, C., Leal-Taixé, L., Sattler, T., Hilsenbeck, S., Cremers, D.: Image-based localization using LSTMs for structured feature correlation (2017). arXiv:1611.07890
Wan, E.A., Van Der Merwe, R.: The unscented kalman filter for nonlinear estimation. In: Proc. IEEE 2000 Adapt. Syst. Signal Proc., Commun., and Control Sympos. (Cat. No.00EX373), pp. 153–158 (2000). https://doi.org/10.1109/ASSPCC.2000.882463
Wang, M., Liu, Q.: A robust cooperative localization algorithm based on covariance intersection method for multi-robot systems. PeerJ Computer Science (2023). https://doi.org/10.7717/peerj-cs.1373
Welch, G., Bishop, G.: An introduction to the kalman filter. Technical Report TR 95-041, University of North Carolina at Chapel Hill, Department of Computer Science (2001). https://www.cs.unc.edu/∼ welch/media/pdf/kalman_intro.pdf
Xiao, L., Wang, J., Qiu, X., Rong, Z., Zou, X.: Dynamic-slam: Semantic monocular visual localization and mapping based on deep learning in dynamic environment. Robot. Auton. Syst. **117**, 1–16 (2019). https://doi.org/10.1016/j.robot.2019.03.012
Yang, M., Huang, C., Huang, X., Hou, S.: Privacy-preserved Visual Simultaneous Localization and Mapping based on a Dual-Component approach. Applied Sciences **15**(5), 2583 (2025). https://doi.org/10.3390/app15052583
Zhang, J., Singh, S.: Loam : Lidar odometry and mapping in real-time, pp. 109–111 (2014)
Zhao, W., Goudar, A., Panerati, J., Schoellig, A.P.: Learning-based bias correction for ultra-wideband localization of resource-constrained Mobile Robots. arXiv preprint arXiv:2003.09371 (2020) arXiv:2003.09371 [cs.RO]
Zou, D., Tan, P., Yu, W.: Collaborative Visual SLAM for multiple agents: A brief survey. Virtual Reality & Intelligent Hardware **1**(5), 461–482 (2019). https://doi.org/10.1016/j.vrih.2019.09.002

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